

Module - 02

Network Layer-Part I

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2.1. IPv4 Addresses

- The **identifier** used in the **IP layer** of the TCP/IP protocol suite to identify the **connection** of each **device** to the **Internet** is called the **Internet address or IP address**.
- An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a host or a router to the Internet.

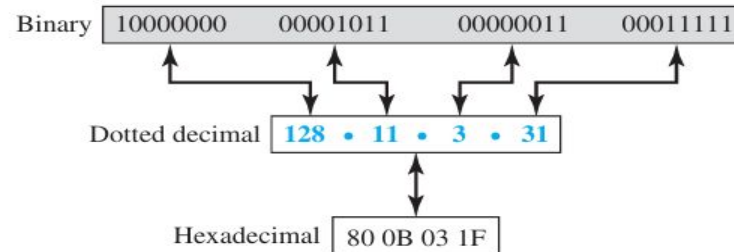
2.1.1. Address Space

- A protocol like IPv4 that defines addresses has an address space. An address space is the total number of addresses used by the protocol.
- If a protocol uses b bits to define an address, the address space is 2^b because each bit can have two different values (0 or 1).
- IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296 (more than four billion).
- If there were no restrictions, more than 4 billion devices could be connected to the Internet.

Notation

- There are three common notations to show an IPv4 address: **binary notation (base 2)**, **dotted-decimal notation (base 256)**, and **hexadecimal notation (base 16)**.

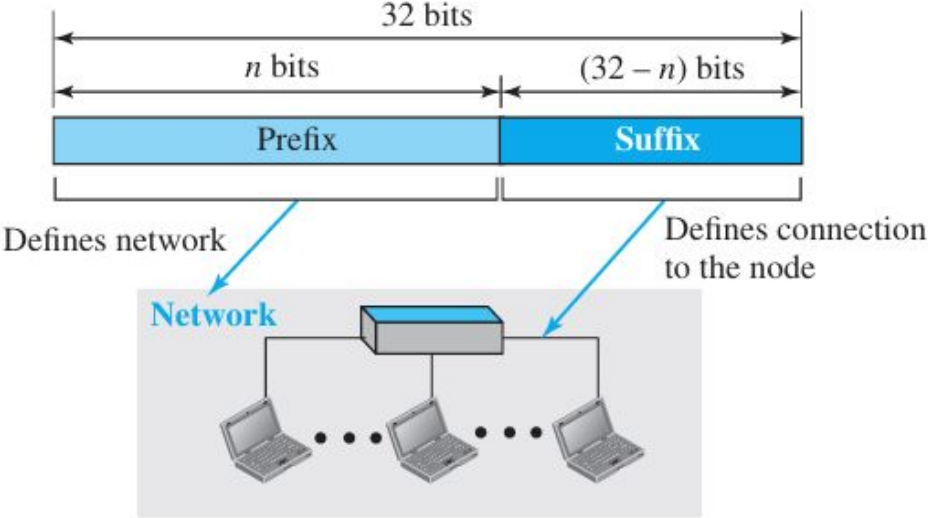
Figure 18.16 Three different notations in IPv4 addressing



Hierarchy in addressing

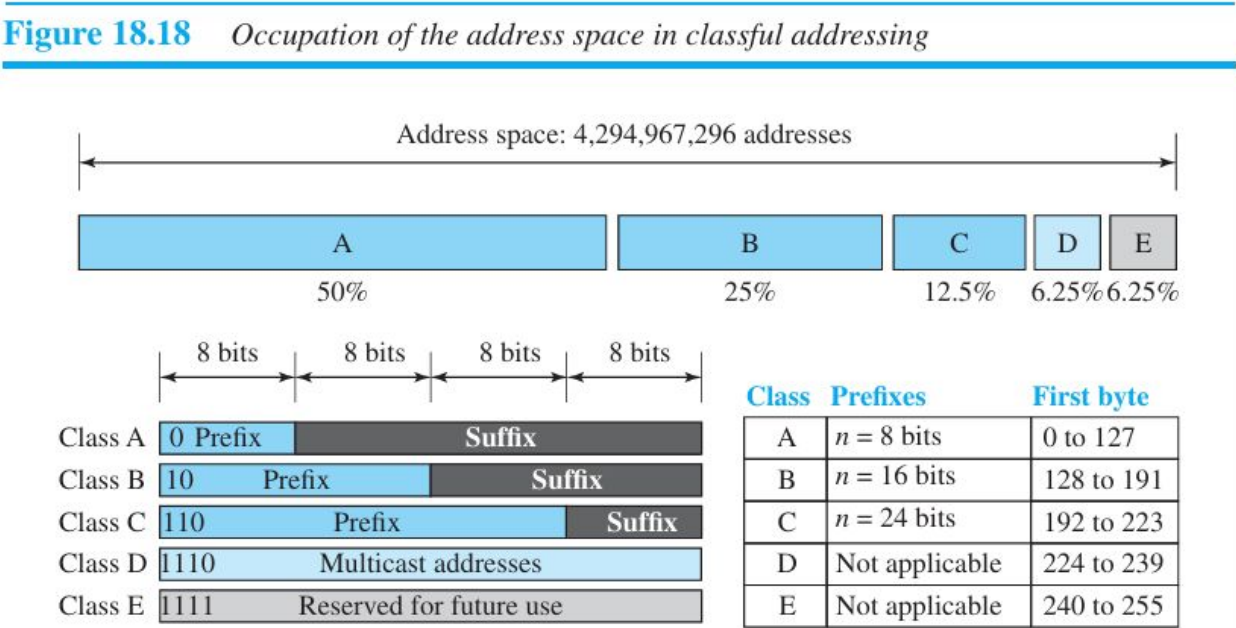
- In any communication network that involves delivery, such as a telephone network or a postal network, the addressing system is hierarchical.

Figure 18.17 *Hierarchy in addressing*



2.1.2. Classful Addressing

- When IPv4 was first created, IP addresses were divided into categories called “Classes” to make it easier to assign addresses to networks of different sizes. **This architecture is called classful Addressing.**
- When the Internet started, an IPv4 address was designed with a fixed-length prefix, but to accommodate both small and large networks, three fixed-length prefixes were designed instead of one ($n = 8$, $n = 16$, and $n = 24$).
- The whole address space was divided into five classes (class A, B, C, D, and E), as shown in **Figure 18.18**.



2.1.2. Classful Addressing

- When IPv4 was first created, IP addresses were divided into categories called “Classes” to make it easier to assign addresses to networks of different sizes. **This architecture is called classful Addressing.**
- When the Internet started, an IPv4 address was designed with a fixed-length prefix, but to accommodate both small and large networks, three fixed-length prefixes were designed instead of one ($n = 8$, $n = 16$, and $n = 24$).
- The whole address space was divided into five classes (class A, B, C, D, and E), as shown in **Figure 18.18**.

Address Depletion

- The reason that classful addressing has become obsolete is address depletion. Since the addresses were not distributed properly, the Internet was faced with the problem of the addresses being rapidly used up, resulting in no more addresses available for organizations and individuals that needed to be connected to the Internet.

Subnetting and Supernetting

- To alleviate address depletion, two strategies were proposed and, to some extent, implemented: **subnetting** and **supernetting**.
- In **subnetting**, a class A or class B block is divided into several subnets. Each subnet has a larger prefix length than the original network.

2.1.3. Classless Addressing

- In classless addressing, the whole address space is divided into variable length blocks. The prefix in an address defines the block (network); the suffix defines the node (device).
- Theoretically, we can have a block of 20, 21, 22, ..., 232 addresses. One of the restrictions, as we discuss later, is that the number of addresses in a block needs to be a power of 2. An organization can be granted one block of addresses.
- Figure 18.19 shows the division of the whole address space into non overlapping blocks.

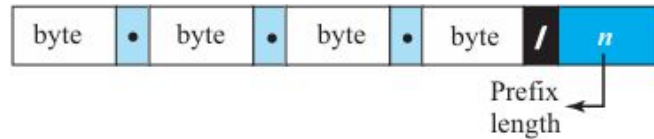
Figure 18.19 *Variable-length blocks in classless addressing*



Prefix Length : Slash Notation

- The first question that we need to answer in classless addressing is how to find the prefix length if an address is given. Since the prefix length is not inherent in the address, we need to separately give the length of the prefix. In this case, the prefix length, n , is added to the address, separated by a slash.
- The notation is informally referred to as slash notation and formally as **classless interdomain routing or CIDR** (pronounced cider) strategy. An address in classless addressing can then be represented as shown in Figure 18.20.

Figure 18.20 *Slash notation (CIDR)*



Examples:

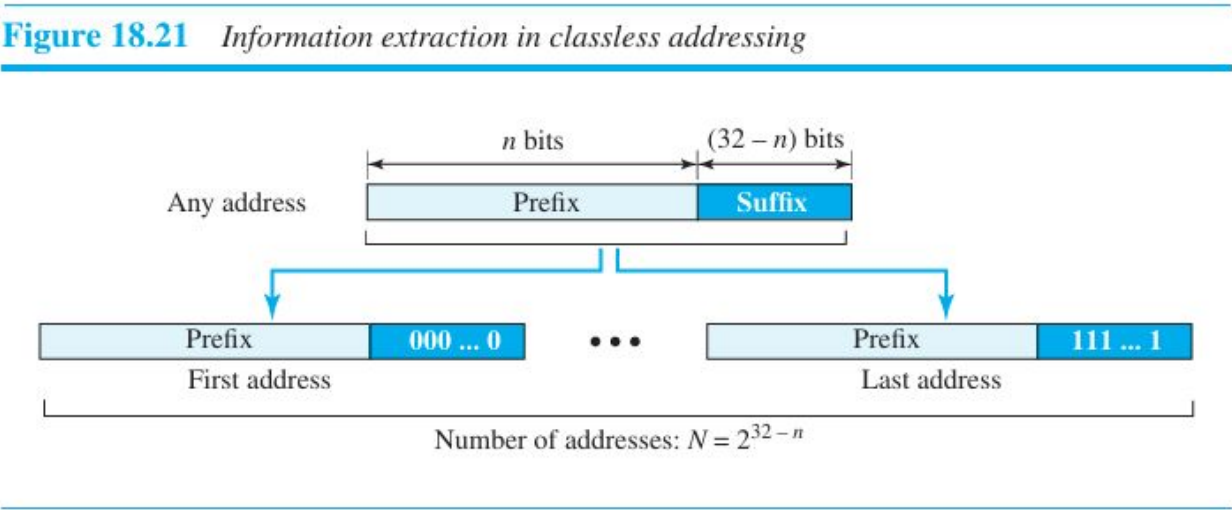
12.24.76.8/8

23.14.67.92/12

220.8.24.255/25

Extracting Information from an Address

- Given any address in the block, we normally like to know three pieces of information about the block to which the address belongs: the number of addresses, the first address in the block, and the last address.
- Since the value of prefix length, n , is given, we can easily find these three pieces of information, as shown in Figure 18.21.



Extracting Information from an Address

Given any address in the block, we normally like to know three pieces of information about the block to which the address belongs: the number of addresses, the first address in the block, and the last address. Since the value of prefix length, n , is given, we can easily find these three pieces of information, as shown in Figure 18.21.

1. The number of addresses in the block is found as $N = 2^{32-n}$.
2. To find the first address, we keep the n leftmost bits and set the $(32 - n)$ rightmost bits all to 0s.
3. To find the last address, we keep the n leftmost bits and set the $(32 - n)$ rightmost bits all to 1s.

Example 1 : A classless address is given as 167.199.170.82/27.

We can find the above three pieces of information as follows.

- The NUMBER OF ADDRESSES in the network is $2^{32-n} = 2^5 = 32$ addresses.
- The FIRST ADDRESS can be found by **keeping the first 27 bits and changing the rest of the bits to 0s.**

Address: **167.199.170.82/27** $\rightarrow$ **10100111 11000111 10101010 01010010**

First address: 167.199.170.64/27 $\rightarrow$ 1**0100111 11000111 10101010 010****00000**

- The LAST ADDRESS can be found by **keeping the first 27 bits and changing the rest of the bits to 1s.**

Address: **167.199.170.82/27** $\rightarrow$ **10100111 11000111 10101010 01010010**

Last address: 167.199.170.95/27 $\rightarrow$ **10100111 11000111 10101010 010****11111**

Example 2 : A classless address is given as 192.168.10.75/26.

Example 3 : A classless address is given as 10.20.30.140/28.

Example 4 : A classless address is given as 172.16.50.200/29.

Example 2 : A classless address is given as 192.168.10.75/26.

We can find the above three pieces of information as follows.

- The **NUMBER OF ADDRESSES** in the network is $2^{32} - n = 2^6 = 64$ addresses.
- The **FIRST ADDRESS** can be found by **keeping the first 26 bits and changing the rest of the bits to 0s.**

Address: 192.168.10.75/26 → 11000000 10101000 00001010 01001011

First address: 192.168.10.64/26 → 11000000 10101000 00001010 01000000

- The **LAST ADDRESS** can be found by **keeping the first 26 bits and changing the rest of the bits to 1s.**

Address: 192.168.10.75/26 → 11000000 10101000 00001010 01001011

Last address: 192.168.10.127/26 → 11000000 10101000 00001010 01111111

Example 3 : A classless address is given as 10.20.30.140/28.

We can find the above three pieces of information as follows.

- The **NUMBER OF ADDRESSES** in the network is $2^{32} - n = 2^4 = 16$ addresses.
- The **FIRST ADDRESS** can be found by **keeping the first 28 bits and changing the rest of the bits to 0s.**

Address: 10.20.30.140/28 → 00001010 00010100 00011110 10001100

First address: 10.20.30.128/28 → 00001010 00010100 00011110 10000000

- The **LAST ADDRESS** can be found by **keeping the first 28 bits and changing the rest of the bits to 1s.**

Address: 10.20.30.140/28 → 00001010 00010100 00011110 10001100

Last address: 10.20.30.143/28 → 00001010 00010100 00011110 10001111

Example 4 : A classless address is given as 172.16.50.200/29.

We can find the above three pieces of information as follows.

- The **NUMBER OF ADDRESSES** in the network is $2^{32-n} = 2^3 = 8$ addresses.
- The **FIRST ADDRESS** can be found by **keeping the first 29 bits and changing the rest of the bits to 0s.**

Address: **172.16.50.200/29** → **10101100 00010000 00110010 11001000**

First address: **172.16.50.200/29** → **10101100 00010000 00110010 11001000**

- The **LAST ADDRESS** can be found by **keeping the first 29 bits and changing the rest of the bits to 1s.**

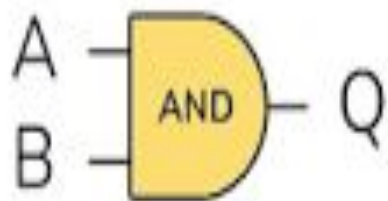
Address: **172.16.50.200/29** → **10101100 00010000 00110010 11001000**

Last address: **172.16.50.207/29** → **10101100 00010000 00110010 11001111**

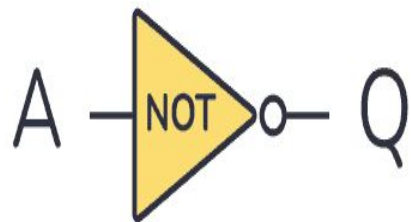
Address Mask

- Another way to find the first and last addresses in the block is to use the address mask. The address mask is a 32-bit number in which the n leftmost bits are set to 1s and the rest of the bits $(32 - n)$ are set to 0s.
- A computer can easily find the address mask because it is the complement of $(2^{32-n} - 1)$.
- The reason for defining a mask in this way is that it can be used by a computer program to extract the information in a block, using the three bitwise operations NOT, AND, and OR.

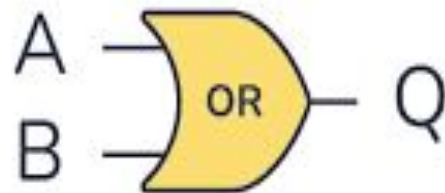
1. **The number of addresses in the block $N = \text{NOT}(\text{mask}) + 1$.**
2. **The first address in the block = (IP address in the block) AND (mask).**
3. **The last address in the block = (IP address in the block) OR [(NOT (mask))].**



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1



A	Q
0	1
1	0



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

Think of /26 as a boundary /26 means: **"The first 26 bits are fixed for identifying the network."**

So the subnet mask marks those **26 bits as 1**.

The remaining 6 bits represent the host portion.

For a subnet mask, **host positions are always written as 0**.

Therefore:

/26
11111111.11111111.11111111.11000000
← ——— 26 ——— → ← 6 →
 network bits host bits

But WHY 0 for host bits?

Because the subnet mask is used to separate network and host portions.

- 1 → this bit belongs to the network
- 0 → this bit belongs to the host

1 = Network

0 = Host

We repeat **167.199.170.82/27** using the mask. The mask in dotted-decimal notation is 255.255.255.224.

The AND, OR, and NOT operations can be applied to individual bytes using calculators and applets at the book website.

Number of addresses in the block: $N = \text{NOT}(\text{mask}) + 1$
 $= 0.0.0.31 + 1$
 $= 32$ addresses

First address: First = (address) AND (mask)
 $= 167.199.170.82$

Last address: Last = (address) OR (NOT mask)
 $= 167.199.170.95$

Explanation **167.199.170.82/27**

27 bits → Network portion

5 bits → Host portion

So the subnet mask in binary is: 11111111.11111111.11111111.11100000

Convert this to decimal: 255 255 255 224

Therefore the Subnet Mask = 255.255.255.224

Number of addresses in the block: $N = \text{NOT}(\text{mask}) + 1$

$= \text{NOT}(11111111.11111111.11111111.11100000) + 1$

$= 00000000.00000000.00000000.00011111 + 1$

$= 0.0.0.31 + 1 \rightarrow \text{Converted to Decimal}$

$= 31 + 1$

$N = 32$

First address: = (address) AND (mask)
 =(167.199.170.82)AND(255.255.255.224)
 = (10100111.11000111.10101010.01010010)
AND (11111111. 11111111. 11111111. 11100000)
 = (10100111.11000111.10101010.01000000)
 = (167 . 199 . 170 . 64)
 = **167.199.170.64 → First Address**

Last address: = (address) OR (NOT(mask))
 =(167.199.170.82)OR(NOT(255.255.255.224))
 =(10100111.11000111.10101010.01010010)OR(NOT(11111111. 11111111. 11111111.11100000))
 =(10100111.11000111.10101010.01010010)OR(00000000.00000000.00000000.00011111)
 = (10100111.11000111.10101010.01010010)
OR (00000000.00000000.00000000.00011111)
 = (10100111.11000111.10101010.01011111)
 = (167 . 199 . 170 . 95)
 = **167.199.170.95 → Last Address**

Example 2 : 192.168.10.75/26 using the mask. The mask in dotted-decimal notation is 255.255.255.192.

Example 3 : 172.16.50.140/28 using the mask. The mask in dotted-decimal notation is 255.255.255.240.

Example 2 : We repeat 192.168.10.75/26 using the mask. The mask in dotted-decimal notation is 255.255.255.192.

**Number of addresses in the block: $N = \text{NOT (mask)} + 1$
= $0.0.0.63 + 1$
= 64 addresses**

**First address: First = (address) AND (mask)
= 192.168.10.64**

**Last address: Last = (address) OR (NOT mask)
= 192.168.10.127**

Example 3 : We repeat 172.16.50.140/28 using the mask. The mask in dotted-decimal notation is 255.255.255.240.

**Number of addresses in the block: $N = \text{NOT (mask)} + 1$
= 0.0.0.15 + 1
= 16 addresses**

**First address: First = (address) AND (mask)
= 172.16.50.128**

**Last address: Last = (address) OR (NOT mask)
= 172.16.50.143**

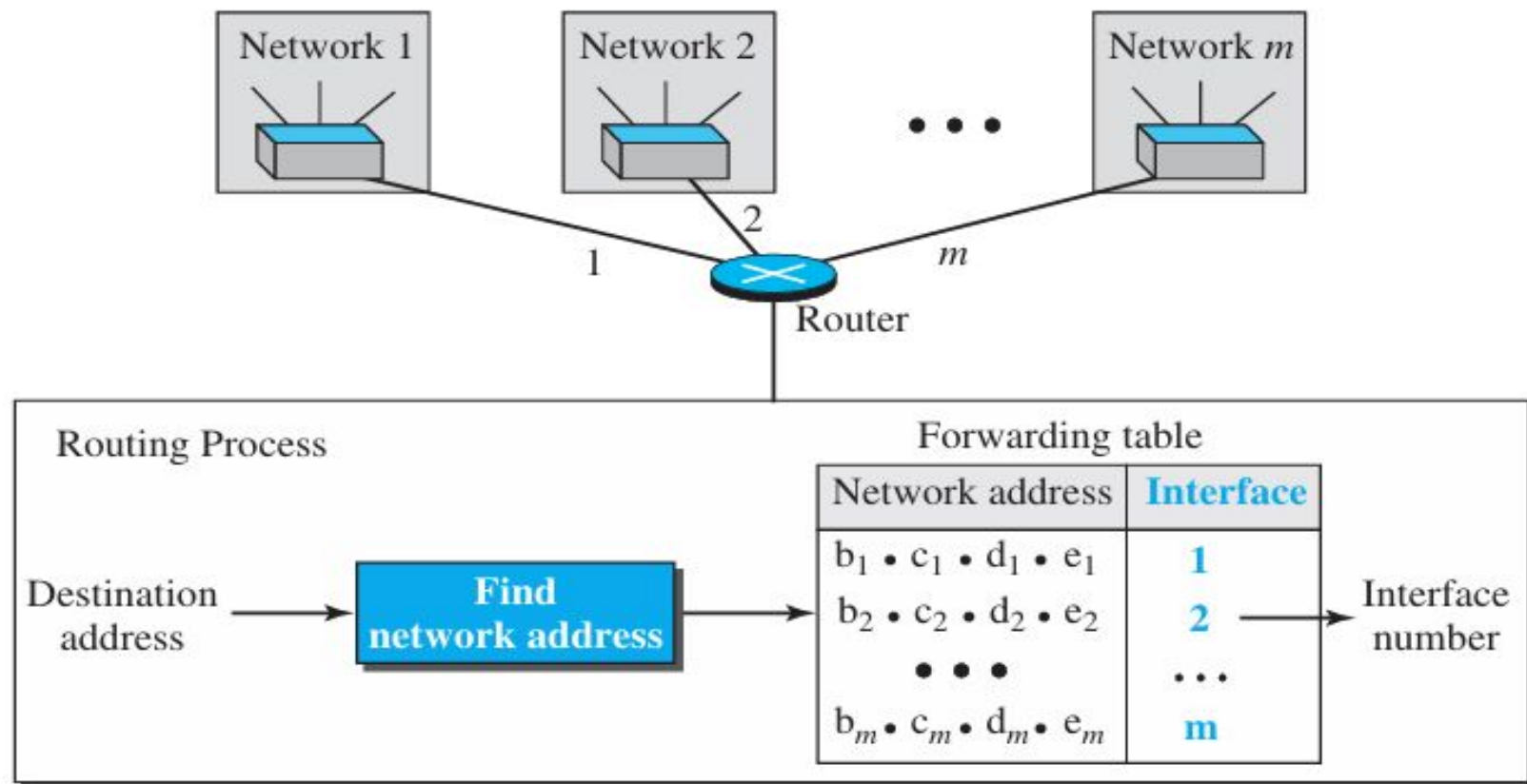
In classless addressing, an address cannot per se define the block the address belongs to. For example, the address 230.8.24.56 can belong to many blocks. Some of them are shown below with the value of the prefix associated with that block.

Prefix length:16	→	Block:	230.8.0.0	to	230.8.255.255
Prefix length:20	→	Block:	230.8.16.0	to	230.8.31.255
Prefix length:26	→	Block:	230.8.24.0	to	230.8.24.63
Prefix length:27	→	Block:	230.8.24.32	to	230.8.24.63
Prefix length:29	→	Block:	230.8.24.56	to	230.8.24.63
Prefix length:31	→	Block:	230.8.24.56	to	230.8.24.57

Network Address

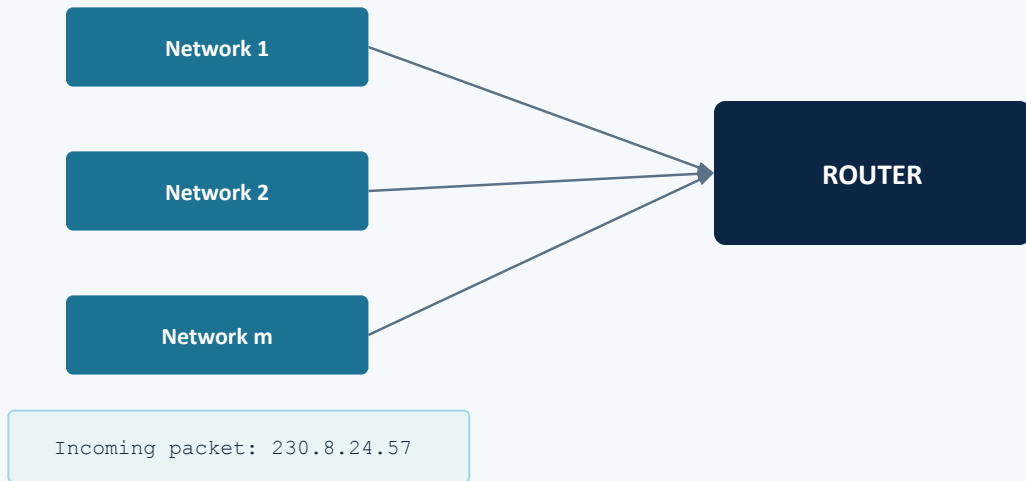
- The above examples show that, given any address, we can find all information about the block. The first address, the network address, is particularly important because it is used in routing a packet to its destination network.
- For the moment, let us assume that an internet is made of m networks and a router with m interfaces. When a packet arrives at the router from any source host, the router needs to know to which network the packet should be sent: from which interface the packet should be sent out.
- When the packet arrives at the network, it reaches its destination host using another strategy that we discuss later.
- After the network address has been found, the router consults its forwarding table to find the corresponding interface from which the packet should be sent out.
- The network address is actually the identifier of the network; each network is identified by its network address.

Figure 18.22 *Network address*



The Network Address — Why Routers Need It

The **first address** in a block is called the **network address**. It acts as the identifier for the whole network — routers use it (not the full destination address) to decide which direction to send a packet.



ROUTER'S FORWARDING TABLE

230.8.0.0/16	iface 1
230.8.16.0/20	iface 2
...	iface ...
230.8.24.32/27	iface m

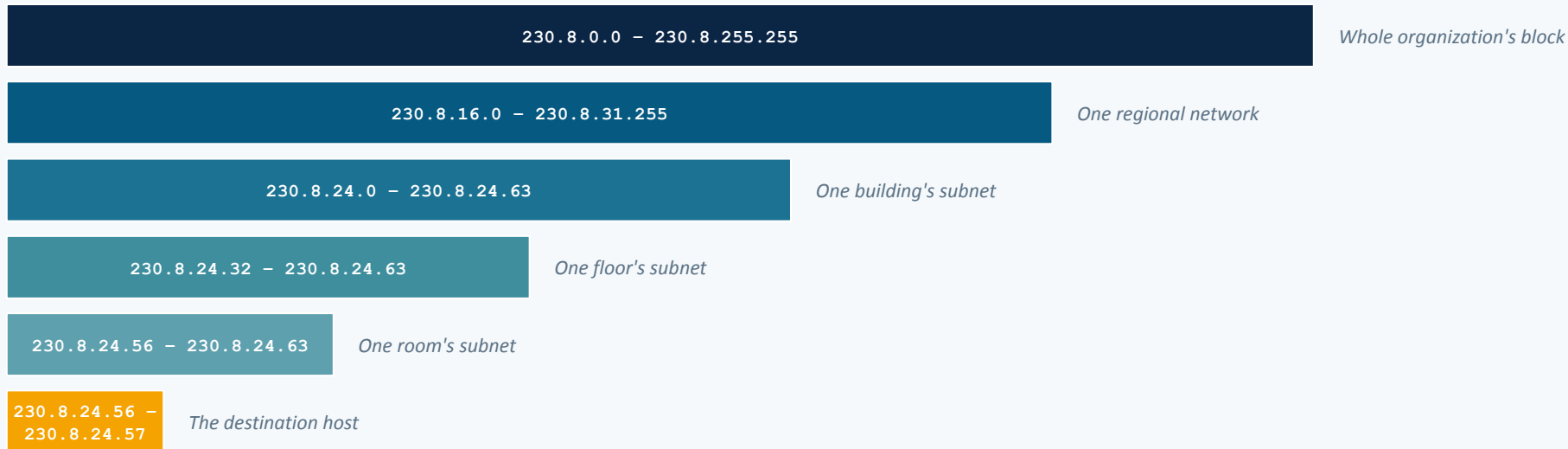
The router strips the packet's destination down to its network address, looks it up, and sends the packet out the matching interface.

💡 REAL-WORLD ANALOGY

A postal sorting center doesn't read your whole address to route a letter across the country — it only reads the city/PIN code. Local delivery figures out the exact street later. A router does the same with the network address.

One Address, Many Levels of Routing

The same idea repeats at every hop: each router only needs enough of the address to know the next narrower range — exactly like Figure 18.22's example.



💡 REAL-WORLD ANALOGY

Just like a delivery narrows down step by step — country → state → city → street → house → person — an IP packet is routed by progressively narrower network addresses until it reaches the exact host.

Block Allocation

- **A block is a group of consecutive IPv4 addresses allocated together to an organization or network.**
- The next issue in classless addressing is block allocation. How are the blocks allocated? The ultimate responsibility of block allocation is given to a global authority called the **Internet Corporation for Assigned Names and Numbers (ICANN)**.
- Example: 192.168.1.0/24 is a block containing **256 consecutive IPv4 addresses**, from 192.168.1.0 to 192.168.1.255.
- ICANN does not normally allocate addresses to individual Internet users. It assigns a large block of addresses to an ISP (or a larger organization that is considered an ISP in this case)(ISP Example: Airtel,Jio, BSNL Etc.,).
- For the proper operation of the CIDR, two restrictions need to be applied to the allocated block.

1. The number of requested addresses, **N, needs to be a power of 2**. The reason is that **$N = 2^{32-n}$ or $n = 32 - \log_2 N$** .

- If N is not a power of 2, we cannot have an integer value for n.

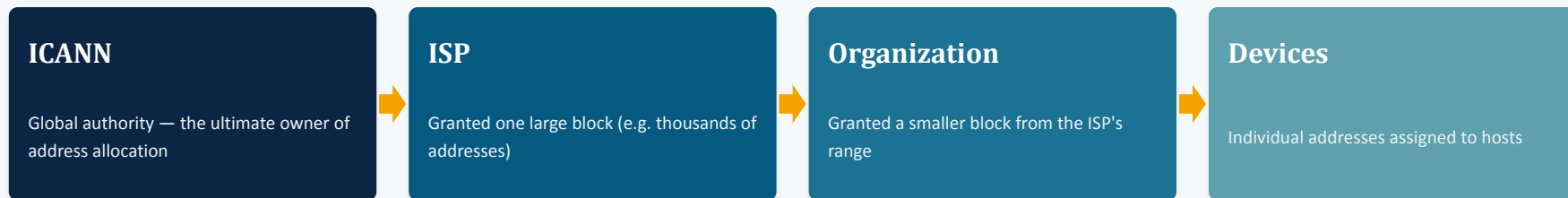
2. The requested block needs to be allocated where there is an adequate number of contiguous addresses available in the address space.

- However, there is a restriction on choosing the first address in the block.
- The first address needs to be divisible by the number of addresses in the block.
- The reason is that the first address needs to be the prefix followed by $(32 - n)$ number of 0s.
- The decimal value of the first address is then

$$\text{first address} = (\text{prefix in decimal}) \times 2^{32-n}$$

$$= (\text{prefix in decimal}) \times N.$$

Who Hands Out Address Blocks?



For the proper operation of CIDR, two restrictions apply to every allocated block:

1 Size must be a power of 2

$$N = 2^{32-n}$$

So $n = 32 - \log_2 N$ always comes out as a whole number. You can't have a block of 1000 addresses — it has to round up to 1024.

2 Start address divisible by N

$$\text{first address} = \text{prefix} \times N$$

The block must begin at an address that's an exact multiple of its own size, so the $(32-n)$ suffix bits are all zero at the start.

An ISP has requested a block of 1000 addresses.

Since 1000 is not a power of 2, 1024 addresses are granted.

The prefix length is calculated as $n = 32 - \log_2 1024 = 22$.

An available block, 18.14.12.0/22, is granted to the ISP.

It can be seen that the first address in decimal is 302,910,464, which is divisible by 1024.

Worked Example: An ISP Requests 1000 Addresses

- 1** Request ISP asks for 1,000 addresses
- 2** Round up to a power of 2 1,024 addresses granted (2^{10})
- 3** Find the prefix length $n = 32 - \log_2(1024) = 32 - 10 = 22$
- 4** Block granted 18.14.12.0/22
- 5** Check rule 2 First address 18.14.12.0 = 302,910,464 in decimal — divisible by 1024 ✓

💡 REAL-WORLD ANALOGY

It's like a phone company asking the government for 1,000 phone numbers. Numbers are only issued in neat blocks (e.g., blocks of 1,024), and the block must start on a clean boundary — not just any random starting number — so the whole numbering system stays organized and easy to route calls through.

Subnetting

- More levels of hierarchy can be created using subnetting. An organization (or an ISP) that is granted a range of addresses may divide the range into several subranges and assign each subrange to a subnetwork (or subnet).
- Note that nothing stops the organization from creating more levels. A subnetwork can be divided into several sub-subnetworks. A sub-subnetwork can be divided into several sub-sub-subnetworks, and so on.

Designing Subnets

- The subnetworks in a network should be carefully designed to enable the routing of packets. We assume the total number of addresses granted to the organization is N , the prefix length is n , the assigned number of addresses to each subnetwork is N_{sub} , and the prefix length for each subnetwork is n_{sub} .

Subnetting — Adding More Levels of Hierarchy

An organization granted a block of addresses can divide that range into smaller **subranges**, one per department or location — called **subnetworks** (or subnets). Nothing stops a subnet from being divided again into sub-subnets, and so on.

THREE RULES FOR DESIGNING SUBNETS

Power of 2

Each subnet's address count must be a power of 2.

Correct prefix

$n_{\text{sub}} = 32 - \lceil \log_2(N_{\text{sub}}) \rceil$ for each subnet.

Largest first

Assign addresses starting with the biggest subnet, so start addresses stay divisible by each subnet's size.



REAL-WORLD ANALOGY

Think of a company that rents one office building with 256 desk slots (a /24 block). Instead of scattering people randomly, the company gives Sales one wing, Engineering another wing, and HR a small corner — each wing (subnet) gets its own contiguous range of desks, sized to how many people actually work there.

Example 1: An organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have 3 sub blocks of addresses to use in its three subnets: one subblock of 10 addresses, one subblock of 60 addresses, and one subblock of 120 addresses. Design the subblocks.

Given block = 14.24.74.0/24

There are 2^{32-n} addresses

$$= 2^{32-24}$$

$$= 2^8$$

=256 addresses

So, **First Address = 14.24.74.0/24**

Last Address = 14.24.74.255/24

Required:

120 addresses(First it should be Power of 2) $\rightarrow$ 128-address block($n_1=32-\log_2 128$) $\rightarrow$ /25

60 addresses(First it should be Power of 2) $\rightarrow$ 64-address block($n_2=32-\log_2 64$) $\rightarrow$ /26

10 addresses(First it should be Power of 2) $\rightarrow$ 16-address block($n_3=32-\log_2 16$) $\rightarrow$ /28

Therefore:

Subblock 1:

14.24.74.0/25

Range: 14.24.74.0 – 14.24.74.127

(First Address) (Last Address)

Subblock 2:

14.24.74.128/26

Range: 14.24.74.128 – 14.24.74.191

(First Address) (Last Address)

Subblock 3:

14.24.74.192/28

Range: 14.24.74.192 – 14.24.74.207

(First Address) (Last Address)

Worked Example: Splitting 14.24.74.0/24

An organization has 14.24.74.0/24 (256 addresses) and needs 3 subnets: 120, 60, and 10 hosts. Allocate largest → smallest.

14.24.74.0/24 — 256 addresses total



Sub-block A (n=25)

needs 120 → rounds up to 128 addresses

14.24.74.0/25 — 14.24.74.127/25

Sub-block B (n=26)

needs 60 → rounds up to 64 addresses

14.24.74.128/26 — 14.24.74.191/26

Sub-block C (n=28)

needs 10 → rounds up to 16 addresses

14.24.74.192/28 — 14.24.74.207/28

Leftover: 128 + 64 + 16 = 208 addresses used, leaving 48 addresses in reserve for future growth — from 14.24.74.208 to 14.24.74.255.

Real-World Example: A Hospital Network

A hospital is granted 192.168.100.0/24 (256 addresses) for three departments: Emergency Room (100 devices), ICU (50 devices), and Admin (20 devices).

Emergency Room

Needs 100 devices

Rounded up to	128
Prefix	/25
First address	192.168.100.0
Last address	192.168.100.127

ICU

Needs 50 devices

Rounded up to	64
Prefix	/26
First address	192.168.100.128
Last address	192.168.100.191

Admin Office

Needs 20 devices

Rounded up to	32
Prefix	/27
First address	192.168.100.192
Last address	192.168.100.223

Example 2. An organization is granted a block of addresses with the beginning address 192.168.10.0/24. The organization needs to have 3 sub blocks of addresses to use in its three subnets: one subblock of 100 addresses, one subblock of 50 addresses, and one subblock of 20 addresses. Design the subblocks.

Example 3. An organization is granted a block of addresses with the beginning address 172.16.20.0/24. The organization needs to have 3 sub blocks of addresses to use in its three subnets: one subblock of 80 addresses, one subblock of 40 addresses, and one subblock of 10 addresses. Design the subblocks.

Example 4. An organization is granted a block of addresses with the beginning address 10.10.5.0/24. The organization needs to have 3 sub blocks of addresses to use in its three subnets: one subblock of 70 addresses, one subblock of 30 addresses, and one subblock of 15 addresses. Design the subblocks.

Example 5. An organization is granted a block of addresses with the beginning address 172.20.10.0/24. The organization needs to have 3 sub blocks of addresses to use in its three subnets: one subblock of 110 addresses, one subblock of 50 addresses, and one subblock of 25 addresses. Design the subblocks.

Example 2: An organization is granted a block of addresses with the beginning address 192.168.10.0/24. The organization needs to have 3 sub blocks of addresses to use in its three subnets: one subblock of 100 addresses, one subblock of 50 addresses, and one subblock of 20 addresses. Design the subblocks.

Given block = 192.168.10.0/24

There are 2^{32-n} addresses
 $= 2^{32-24}$
 $= 2^8$
 $= 256$ addresses

So, First Address = 192.168.10.0/24
Last Address = 192.168.10.255/24

Required:

100 addresses (First it should be Power of 2) $\rightarrow$ 128-address block ($n_1 = 32 - \log_2 128$) $\rightarrow$ /25

50 addresses (First it should be Power of 2) $\rightarrow$ 64-address block ($n_2 = 32 - \log_2 64$) $\rightarrow$ /26

20 addresses (First it should be Power of 2) $\rightarrow$ 32-address block ($n_3 = 32 - \log_2 32$) $\rightarrow$ /27

Therefore:

Subblock 1:

192.168.10.0/25

Range: 192.168.10.0 – 192.168.10.127
(First Address) (Last Address)

Subblock 2:

192.168.10.128/26

Range: 192.168.10.128 – 192.168.10.191
(First Address) (Last Address)

Subblock 3:

192.168.10.192/27

Range: 192.168.10.192 – 192.168.10.223
(First Address) (Last Address)

Example 3: An organization is granted a block of addresses with the beginning address 172.16.20.0/24. The organization needs to have 3 sub blocks of addresses to use in its three subnets: one subblock of 80 addresses, one subblock of 40 addresses, and one subblock of 10 addresses. Design the subblocks.

Given block = 172.16.20.0/24

There are 2^{32-n} addresses
= 2^{32-24}
= 2^8
= 256 addresses

So, First Address = 172.16.20.0/24
Last Address = 172.16.20.255/24

Required:

80 addresses (First it should be Power of 2) → 128-address block ($n_1 = 32 - \log_2 128$) → /25

40 addresses (First it should be Power of 2) → 64-address block ($n_2 = 32 - \log_2 64$) → /26

10 addresses (First it should be Power of 2) → 16-address block ($n_3 = 32 - \log_2 16$) → /28

Therefore:

Subblock 1:

172.16.20.0/25

Range: 172.16.20.0 – 172.16.20.127

(First Address) (Last Address)

Subblock 2:

172.16.20.128/26

Range: 172.16.20.128 – 172.16.20.191

(First Address) (Last Address)

Subblock 3:

172.16.20.192/28

Range: 172.16.20.192 – 172.16.20.207

(First Address) (Last Address)

Example 4: An organization is granted a block of addresses with the beginning address 10.10.5.0/24. The organization needs to have 3 sub blocks of addresses to use in its three subnets: one subblock of 70 addresses, one subblock of 30 addresses, and one subblock of 15 addresses. Design the subblocks.

Given block = 10.10.5.0/24

There are 2^{32-n} addresses
 $= 2^{32-24}$
 $= 2^8$
 $= 256$ addresses

So, First Address = 10.10.5.0/24
Last Address = 10.10.5.255/24

Required:

70 addresses (First it should be Power of 2) $\rightarrow$ 128-address block ($n1 = 32 - \log_2 128$) $\rightarrow$ /25

30 addresses (First it should be Power of 2) $\rightarrow$ 32-address block ($n2 = 32 - \log_2 32$) $\rightarrow$ /27

15 addresses (First it should be Power of 2) $\rightarrow$ 16-address block ($n3 = 32 - \log_2 16$) $\rightarrow$ /28

Therefore:

Subblock 1:

10.10.5.0/25

Range: 10.10.5.0 – 10.10.5.127

(First Address) (Last Address)

Subblock 2:

10.10.5.128/27

Range: 10.10.5.128 – 10.10.5.159

(First Address) (Last Address)

Subblock 3:

10.10.5.160/28

Range: 10.10.5.160 – 10.10.5.175

(First Address) (Last Address)

Example 5: An organization is granted a block of addresses with the beginning address 172.20.10.0/24. The organization needs to have 3 sub blocks of addresses to use in its three subnets: one subblock of 110 addresses, one subblock of 50 addresses, and one subblock of 25 addresses. Design the subblocks.

Given block = 172.20.10.0/24

There are 2^{32-n} addresses
= 2^{32-24}
= 2^8
= 256 addresses

So, First Address = 172.20.10.0/24
Last Address = 172.20.10.255/24

Required:

110 addresses (First it should be Power of 2) →
128-address block ($n_1 = 32 - \log_2 128$) → /25

50 addresses (First it should be Power of 2) →
64-address block ($n_2 = 32 - \log_2 64$) → /26

25 addresses (First it should be Power of 2) →
32-address block ($n_3 = 32 - \log_2 32$) → /27

Therefore:

Subblock 1:

172.20.10.0/25

Range: 172.20.10.0 – 172.20.10.127

(First Address) (Last Address)

Subblock 2:

172.20.10.128/26

Range: 172.20.10.128 – 172.20.10.191

(First Address) (Last Address)

Subblock 3:

172.20.10.192/27

Range: 172.20.10.192 – 172.20.10.223

(First Address) (Last Address)

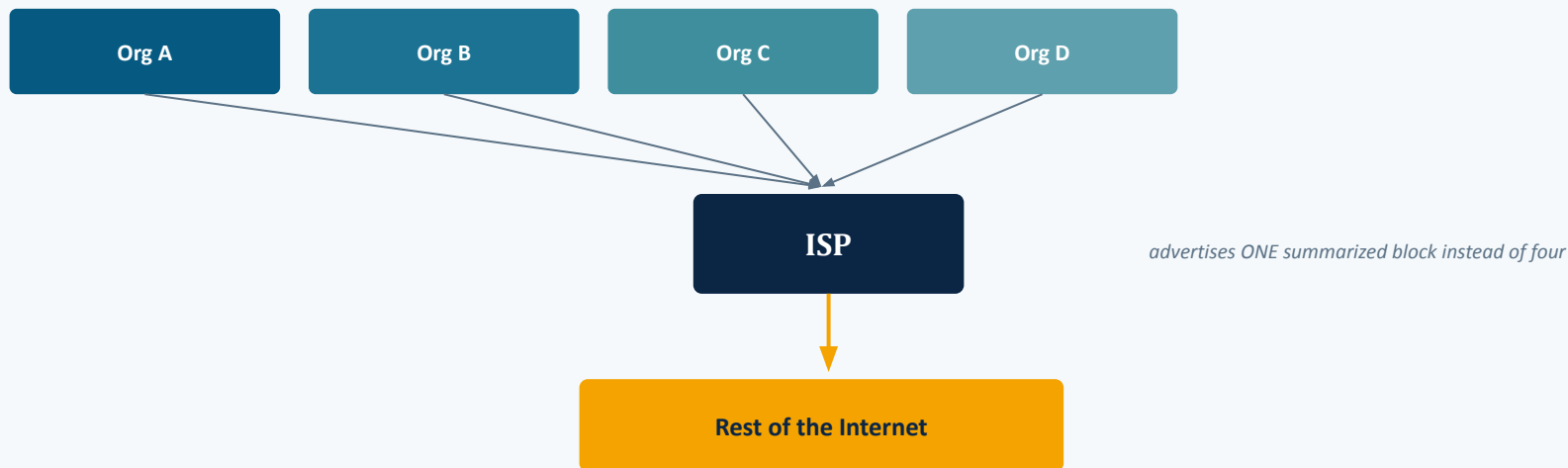
1. An organization is assigned the address block 192.168.40.0/24. It has three departments requiring approximately 55, 110, and 25 addresses respectively. The network administrator wants to minimize address wastage and allocate the addresses using VLSM. Design suitable subblocks for all three departments and specify the network address, prefix length, and address range of each subblock.
2. A company receives the block 172.18.0.0/24. Three branches require 90, 45, and 12 addresses respectively. The administrator decides to allocate the largest requirement first and then continue with the remaining address space. Determine the VLSM allocation and identify the unused address range after all three subblocks are assigned.
3. A college is given the address block 10.20.30.0/24. The Computer Science department requires 100 addresses, the ECE department requires 50 addresses, and the Administration department requires 20 addresses. Design the subblocks using VLSM. For each subblock, determine the prefix length, first address, and last address.
4. An organization has been assigned 172.25.16.0/24. Its three networks require 70, 30, and 14 addresses. The administrator wants every requirement to be satisfied using the smallest possible power-of-two block. Determine the required block size for each network and design the complete VLSM allocation.
5. A company owns the address block 192.168.50.0/24. It wants to create three subnets for different purposes. One subnet must support at least 115 addresses, another must support at least 55 addresses, and the third must support at least 25 addresses. Determine whether all three requirements can be accommodated within the given /24 block. If possible

Address Aggregation

- One of the advantages of the CIDR strategy is address aggregation (sometimes called address summarization or route summarization). When blocks of addresses are combined to create a larger block, routing can be done based on the prefix of the larger block.
- ICANN assigns a large block of addresses to an ISP. Each ISP in turn divides its assigned block into smaller subblocks and grants the subblocks to its customers.

Address Aggregation — Combining Blocks

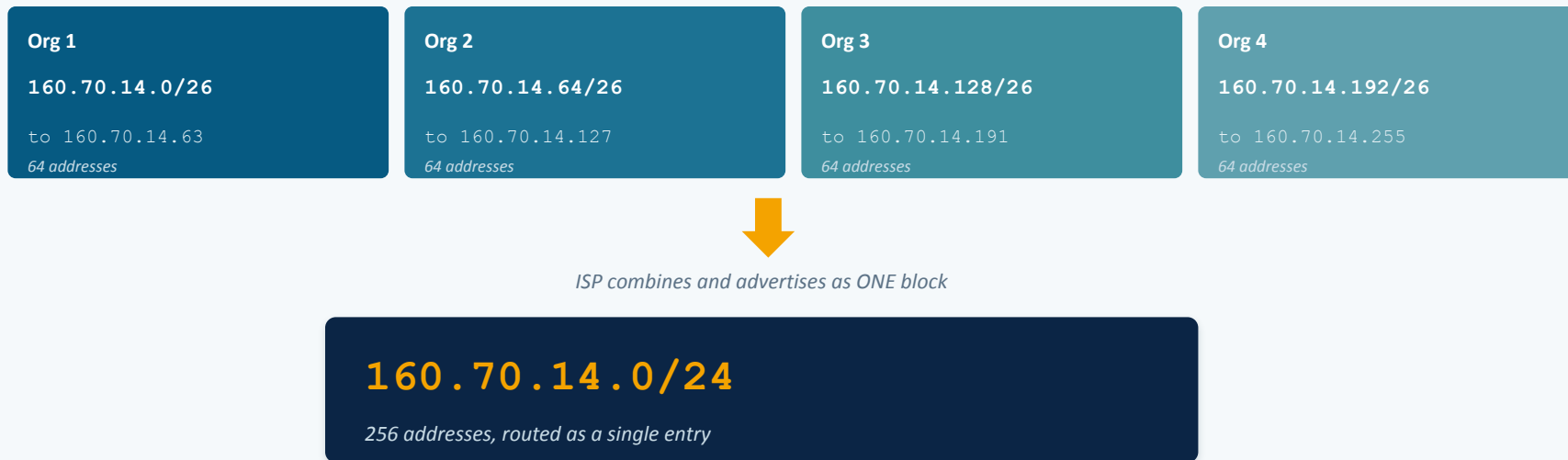
Aggregation (also called *route summarization*) is the reverse of subnetting: several small blocks are combined so routing can happen using just the **prefix of the larger block**. This is one of CIDR's biggest advantages — far fewer entries in the world's routing tables.



💡 REAL-WORLD ANALOGY

A regional shipping hub doesn't tell the national depot every individual customer address in its city — it just says “send anything for this ZIP code range to me,” then sorts deliveries locally. The ISP does the same with routing.

Worked Example: Four Blocks Become One



How it works: Any packet destined for 160.70.14.0 – 160.70.14.255 is sent to this ISP by the rest of the Internet. It's then the ISP's job to forward it internally to the correct organization — just like international mail is sent to a country's capital first, then redistributed locally.

Five Special-Purpose IPv4 Addresses

A handful of address blocks are reserved for special jobs rather than identifying an ordinary host on the public Internet.

Name	Block	What it's for
This-host	0.0.0.0/32	"I don't know my address yet" — used when a host has no address of its own.
Limited-broadcast	255.255.255.255/32	Send to every device on THIS network only — routers block it from leaving.
Loopback	127.0.0.0/8	Talk to yourself — tests software on the same machine, never leaves the host.
Private	10.0.0.0/8 + 3 more	Used inside private networks only; not routable on the public Internet.
Multicast	224.0.0.0/4	One-to-many delivery — only subscribed devices receive it.

This-Host & Limited-Broadcast

0.0.0.0/32 — This-Host

Used when a device needs to send a packet but doesn't yet know its own IP address — for example, right when it's first booting up and asking for an address.

REAL-WORLD ANALOGY

It's like writing "FROM: New Employee (no badge yet)" on an internal envelope on your first day — you don't have an ID number yet, so you use a placeholder until HR assigns you one.

Where you'll see it: the very first message a device sends during DHCP (covered later in this deck) often uses 0.0.0.0 as its source address.

255.255.255.255/32

Limited-Broadcast

Used to send a message to every device on the local network at once. Routers refuse to forward it any further — it dies at the edge of this one network.

REAL-WORLD ANALOGY

It's the office PA system: "Attention all staff on this floor..." Everyone in the building hears it, but the announcement doesn't travel to the branch office across town.

Where you'll see it: DHCP servers reply to a brand-new device using this address, because the device doesn't have a real IP to receive a direct reply yet.

Loopback & Private Addresses

127.0.0.0/8 — Loopback

Any address in this block (commonly 127.0.0.1) loops back to the same machine — the packet never actually leaves the computer. Used to test client/server software locally before deploying across real machines.

ANALOGY

Like talking to yourself in a mirror — the message never leaves the room, but it's a perfectly good way to rehearse.

Private Addresses — for internal use only

These four blocks can be reused by any home or office network. They are never routed on the public Internet — a router called NAT translates them to a public address at the edge.

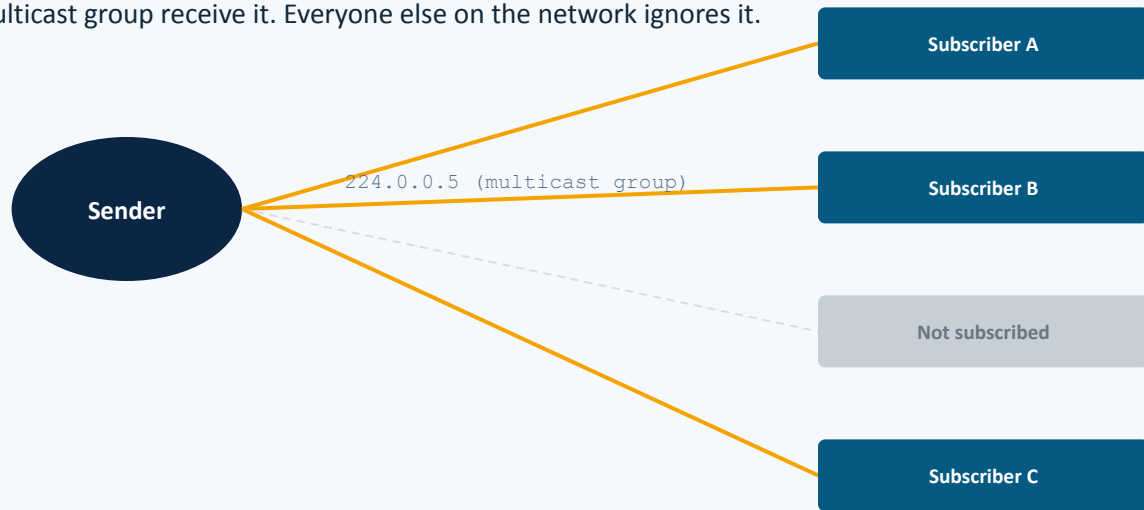
10.0.0.0/8	Large organizations
172.16.0.0/12	Medium networks
192.168.0.0/16	Home & small-office routers
169.254.0.0/16	Auto-assigned when DHCP fails (APIPA)

REAL-WORLD ANALOGY

Just like internal office extensions (dial “101” for reception) only work inside that one building — and every office building in the world can reuse “101” without conflict — private IP addresses are reused inside millions of separate home and office networks. To call from outside, you need the building's public phone number (its public IP), which NAT provides.

Multicast Addresses — 224.0.0.0/4

Multicast lets one sender deliver a single stream of packets to **many receivers at once** — but only the devices that have subscribed to that multicast group receive it. Everyone else on the network ignores it.



Only devices that joined the multicast group (dashed amber lines) receive the stream — unsubscribed devices (gray) never see the traffic.

💡 REAL-WORLD ANALOGY

It's exactly like a radio station broadcasting on 101.5 FM: anyone with a radio tuned to that frequency receives the show. People with radios tuned elsewhere — or no radio at all — never hear it, even though the signal passes right by them.

Special Addresses

- Before finishing the topic of addresses in IPv4, we need to mention five special addresses that are used for special purposes: *this-host address, limited-broadcast address, loopback address, private addresses, and multicast addresses.*

2.1.4. DHCP and NAT

- We have seen that a large organization or an ISP can receive a block of addresses directly from ICANN and a small organization can receive a block of addresses from an ISP.
- After a block of addresses are assigned to an organization, the network administrator can manually assign addresses to the individual hosts or routers.
- However, address assignment in an organization can be done automatically using the **Dynamic Host Configuration Protocol (DHCP).**
- **DHCP is a network management protocol used to dynamically assign an IP address to many devices/ nodes on a network so they can communicate using IP.**

DHCP Message Format

- DHCP is a client-server protocol in which the client sends a request message and the server returns a response message. Before we discuss the operation of DHCP, let us show the general format of the DHCP message in Figure 18.25.

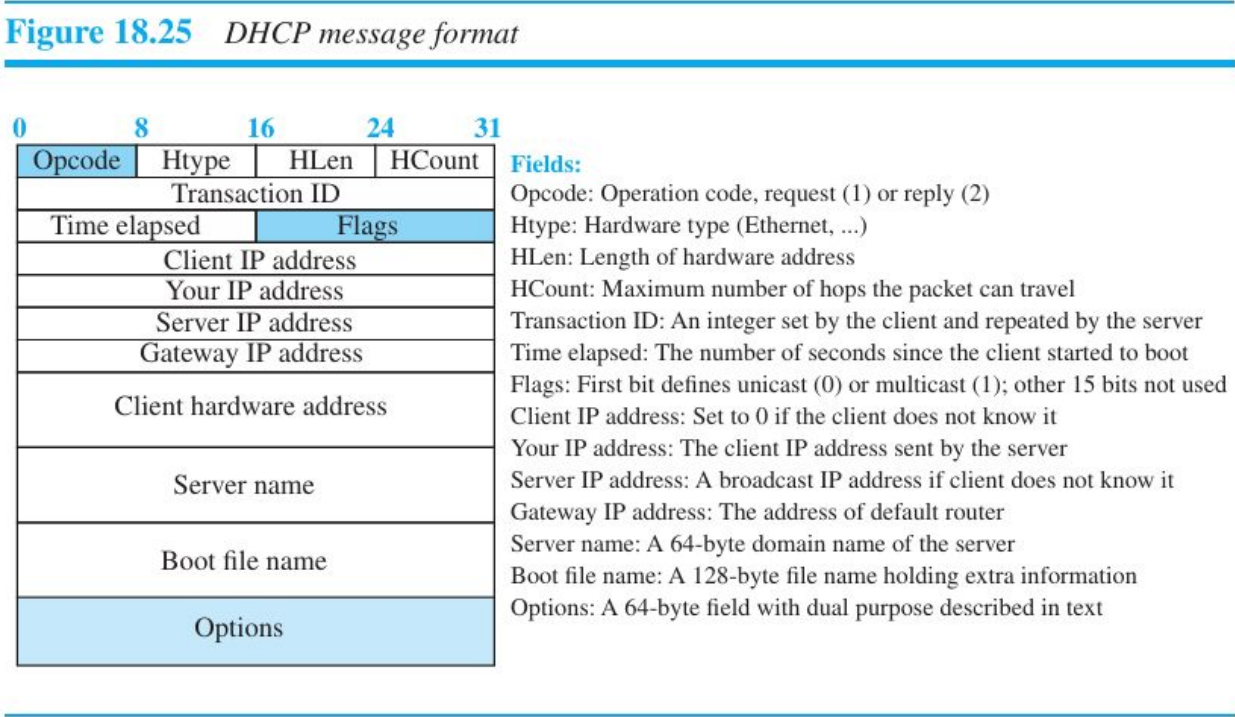


Figure 18.26 *Option format*

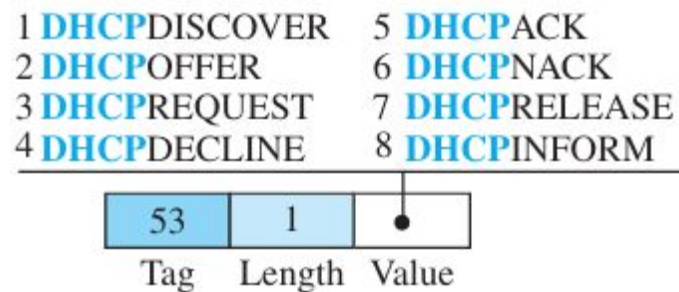
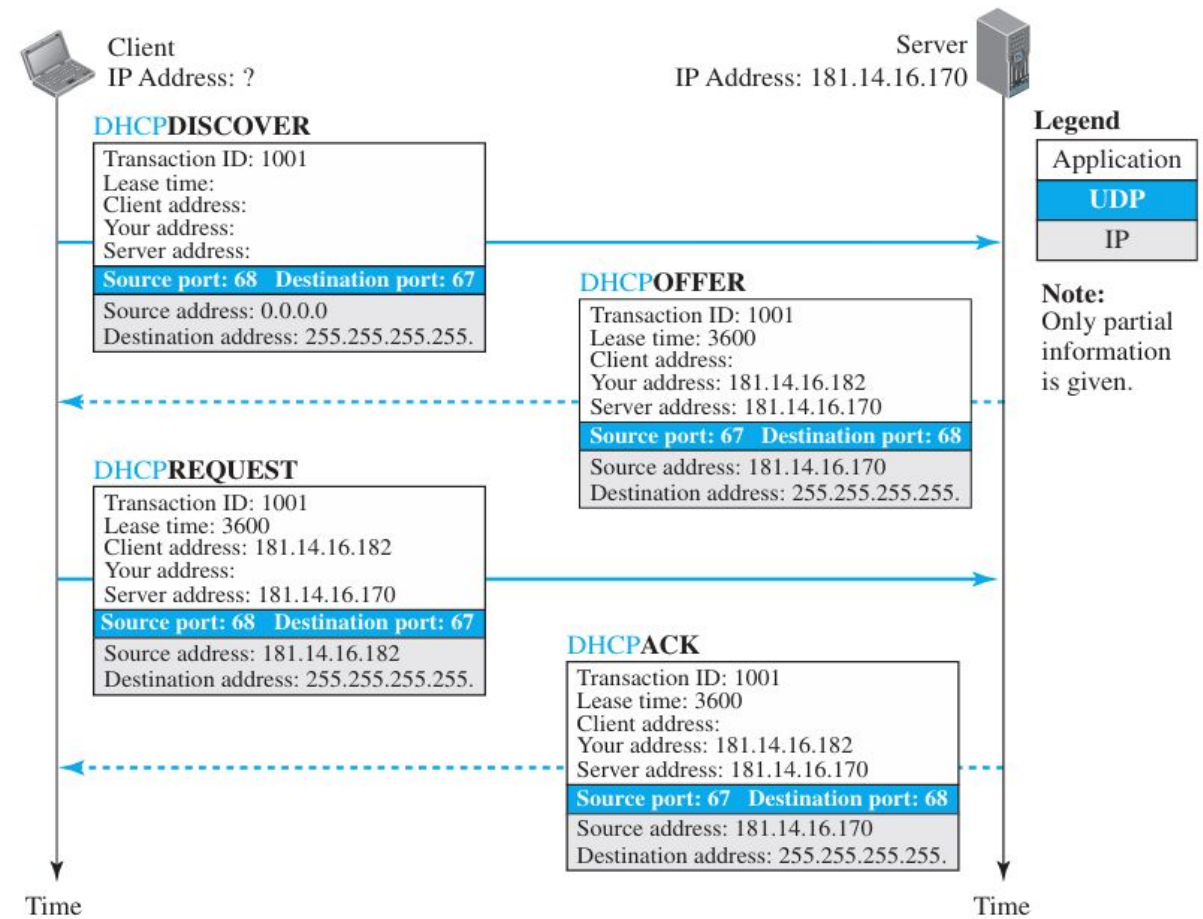


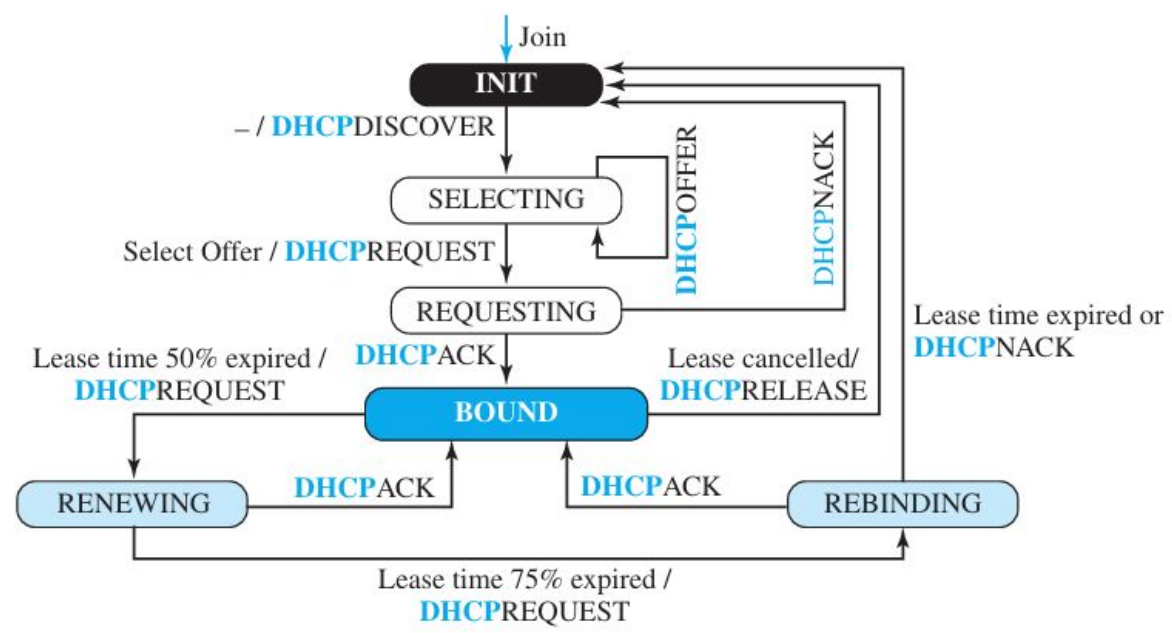
Figure 18.27 Operation of DHCP



Transition States

- The previous scenarios we discussed for the operation of the DHCP were very simple. To provide dynamic address allocation, the DHCP client acts as a state machine that performs transitions from one state to another depending on the messages it receives or sends.
- Figure 18.28 shows the transition diagram with the main states.

Figure 18.28 *FSM for the DHCP client*

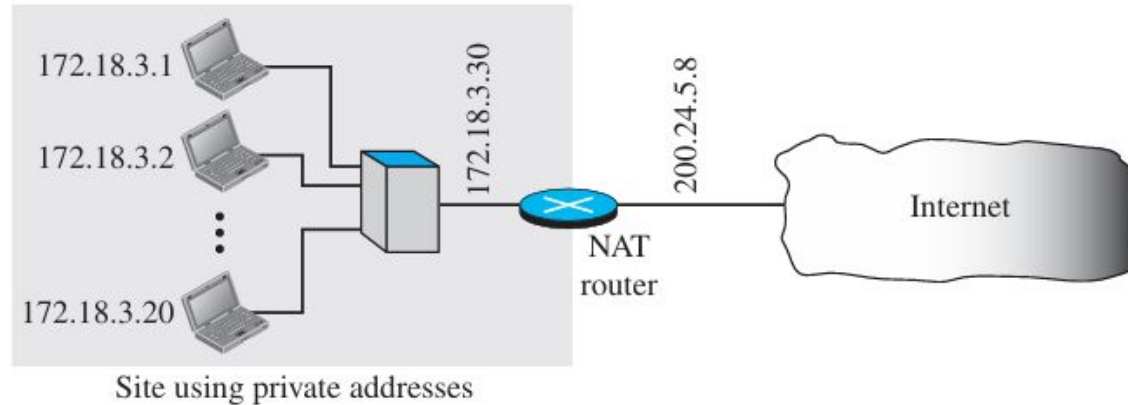


- When the DHCP client first starts, it is in the INIT state (initializing state). The client broadcasts a discover message. When it receives an offer, the client goes to the SELECTING state.
- While it is there, it may receive more offers. After it selects an offer, it sends a request message and goes to the REQUESTING state.
- If an ACK arrives while the client is in this state, it goes to the BOUND state and uses the IP address.
- When the lease is 50 percent expired, the client tries to renew it by moving to the RENEWING state.
- If the server renews the lease, the client moves to the BOUND state again.
- If the lease is not renewed and the lease time is 75 percent expired, the client moves to the REBINDING state.
- If the server agrees with the lease (ACK message arrives), the client moves to the BOUND state and continues using the IP address;
- otherwise, the client moves to the INIT state and requests another IP address.
- Note that the client can use the IP address only when it is in the BOUND, RENEWING, or REBINDING state.
- The above procedure requires that the client uses three timers: renewal timer (set to 50 percent of the lease time), rebinding timer (set to 75 percent of the lease time), and expiration timer (set to the lease time).

NAT(Network Address Translation)

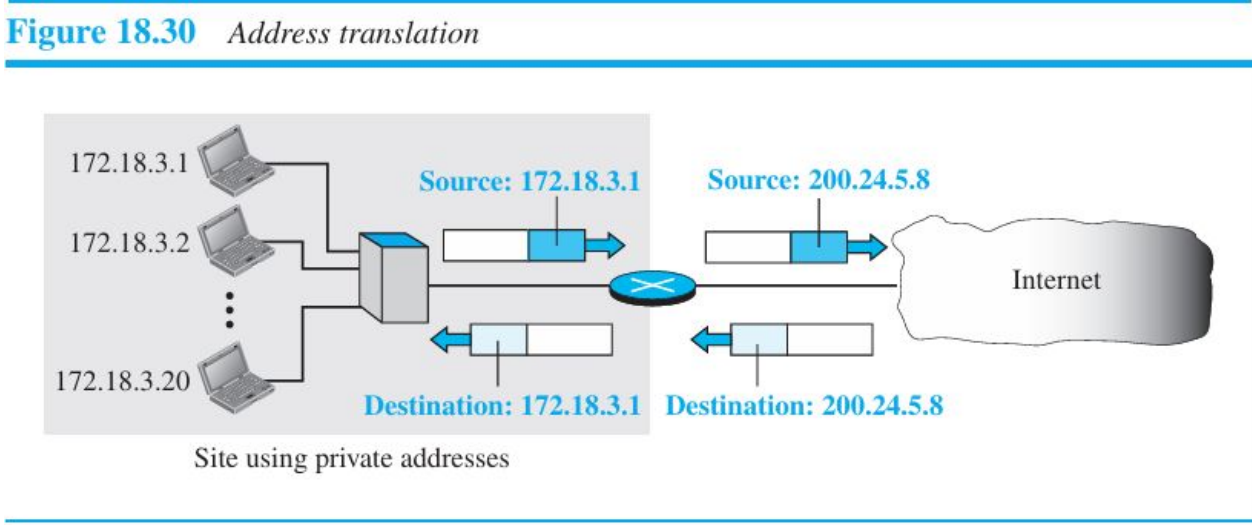
- The technology allows a site to use a set of private addresses for internal communication and a set of global Internet addresses (at least one) for communication with the rest of the world. The site must have only one connection to the global Internet through a NAT-capable router that runs NAT software.
- Figure 18.29 shows a simple implementation of NAT.

Figure 18.29 NAT



Address translation

- All of the outgoing packets go through the NAT router, which replaces the source address in the packet with the global NAT address.
- All incoming packets also pass through the NAT router, which replaces the destination address in the packet (the NAT router global address) with the appropriate private address.
- Figure 18.30 shows an example of address translation.

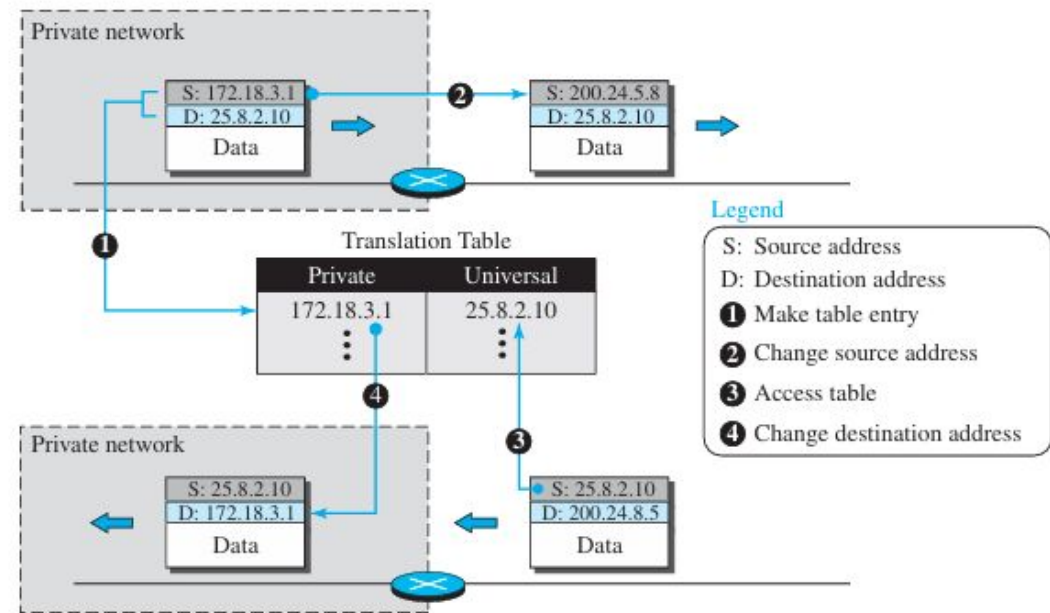


Translation table

Using One IP Address

- In its simplest form, a translation table has only two columns: the private address and the external address (destination address of the packet).
- When the router translates the source address of the outgoing packet, it also makes note of the destination address— where the packet is going.
- When the response comes back from the destination, the router uses the source address of the packet (as the external address) to find the private address of the packet.
- Figure 18.31 shows the idea.

Figure 18.31 Translation



Using the Pool of IP addresses

- The use of only one global address by the NAT router allows only one private-network host to access a given external host.
- To remove this restriction, the NAT router can use a pool of global addresses. For example, instead of using only one global address (200.24.5.8), the NAT router can use four addresses (200.24.5.8, 200.24.5.9, 200.24.5.10, and 200.24.5.11).

Using Both IP Addresses and Port Addresses

- To allow a many-to-many relationship between private-network hosts and external server programs, we need more information in the translation table.

2.2.Forwarding of IP Packets

- As we discussed before, forwarding means to place the packet in its route to its destination.
- Since the Internet today is made of a combination of links (networks), forwarding means to deliver the packet to the next hop (which can be the final destination or the intermediate connecting device).

2.2.1. Forwarding based on destination address

- In forwarding based on destination address, forwarding requires a host or router to have a forwarding table.
- When a host has a packet to send or when a router has received a packet to be forwarded, it looks at this table to find the next hop to deliver the packet to.
- When data is sent over the internet, it travels in **packets**. Each packet has a **destination IP address**, and routers need to look at this address to figure out **where to send the packet next**.

Figure 18.32 *Simplified forwarding module in classless address*

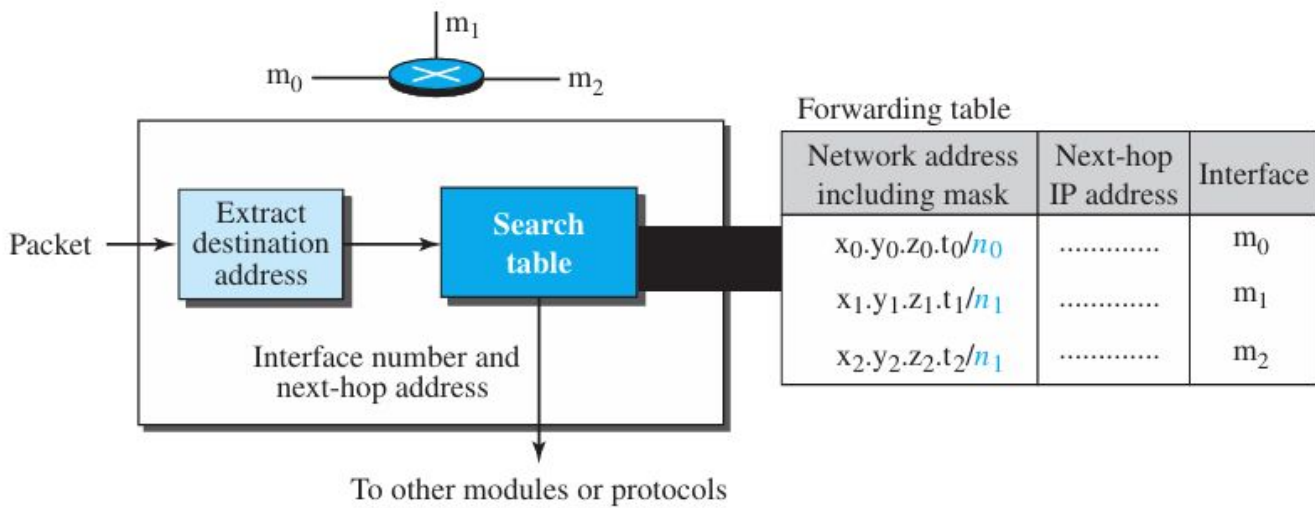


Figure 18.33 Configuration for Example 18.7

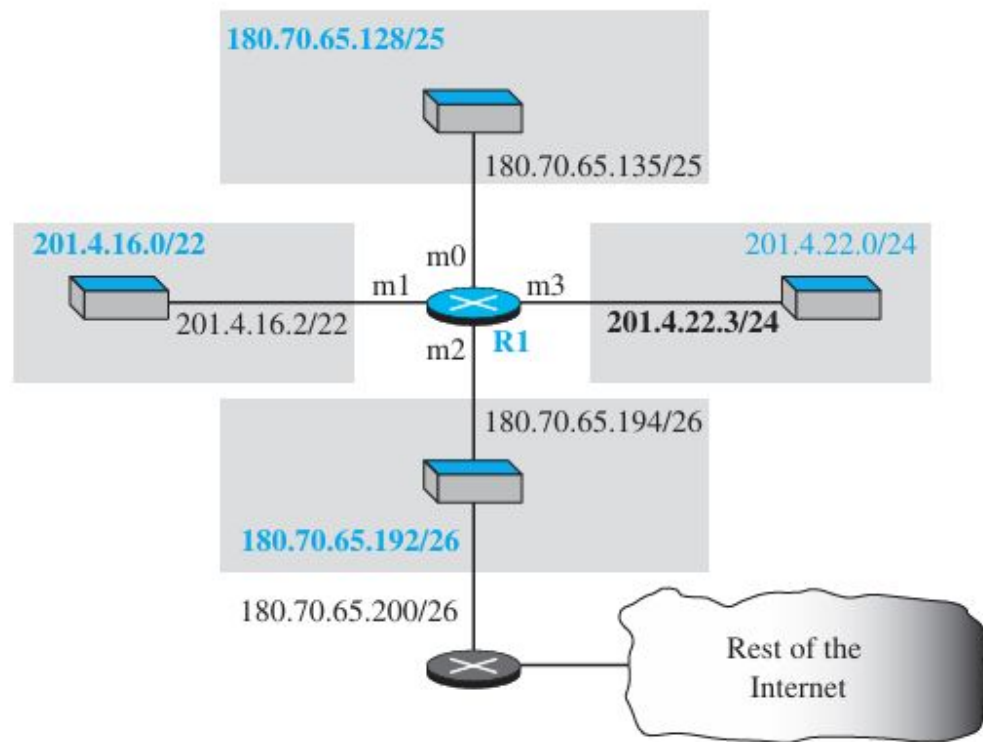


Table 18.2 *Forwarding table for router R1 in Figure 18.33*

<i>Network address/mask</i>	<i>Next hop</i>	<i>Interface</i>
180.70.65.192/ 26	—	m2
180.70.65.128/ 25	—	m0
201.4.22.0/ 24	—	m3
201.4.16.0/ 22	—	m1
Default	180.70.65.200	m2

- Instead of Table 18.2, we can use Table 18.3, in which the network address/mask is given in bits.

Table 18.3 Forwarding table for router R1 in Figure 18.33 using prefix bits

<i>Leftmost bits in the destination address</i>	<i>Next hop</i>	<i>Interface</i>
10110100 01000110 01000001 11	—	m2
10110100 01000110 01000001 1	—	m0
11001001 00000100 00011100	—	m3
11001001 00000100 000100	—	m1
Default	180.70.65.200	m2

- When a packet arrives whose leftmost 26 bits in the destination address match the bits in the first row, the packet is sent out from interface m2.
- When a packet arrives whose leftmost 25 bits in the address match the bits in the second row, the packet is sent out from interface m0, and so on.
- The table clearly shows that the first row has the longest prefix and the fourth row has the shortest prefix.
- The longer prefix means a smaller range of addresses; the shorter prefix means a larger range of addresses.

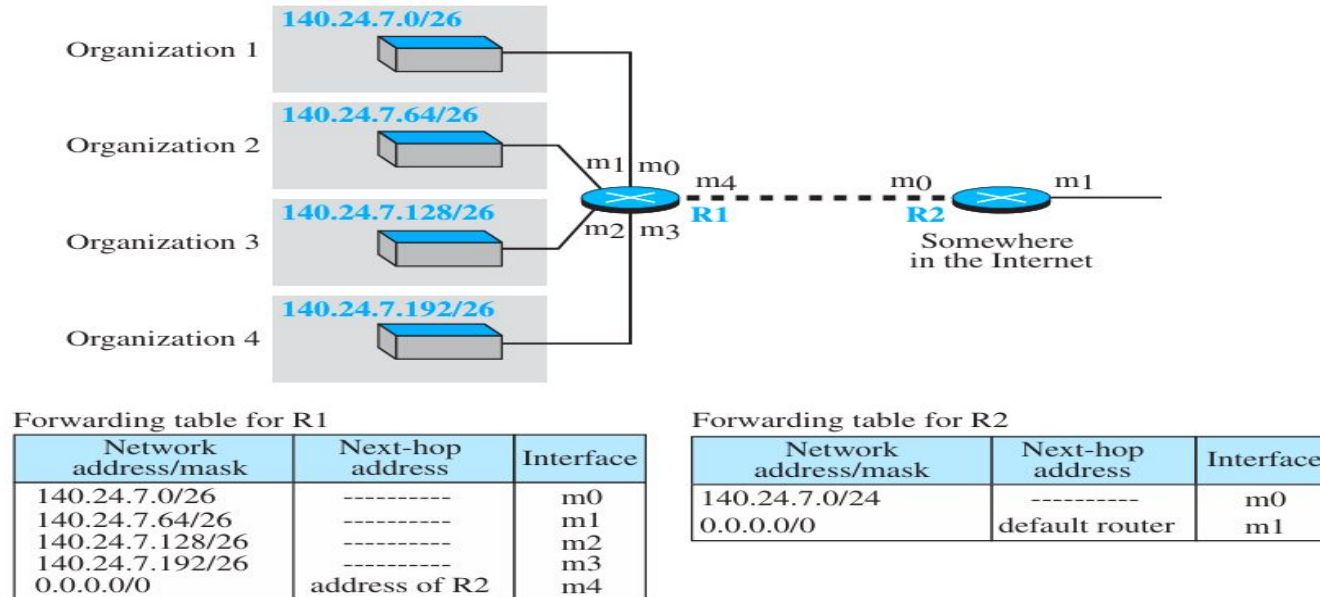
Address aggregation

- **Imagine This**
- There are **4 small companies**. Each one gets **64 IP addresses**:
 - Company A: 140.24.7.0 – 140.24.7.63
 - Company B: 140.24.7.64 – 140.24.7.127
 - Company C: 140.24.7.128 – 140.24.7.191
 - Company D: 140.24.7.192 – 140.24.7.255
- That's **4 blocks of 64 addresses**, all next to each other.
- **Router R1 (Close to the companies)** R1 is connected directly to the 4 companies. It **needs to know exactly** which block goes to which company
- So R1 has **4 entries** in its table (one for each block).
- **Router R2 (Far away)** R2 is **not directly connected** to the companies.
- All 4 blocks fall within a bigger block: 140.24.7.0 to 140.24.7.255 → That's **256 addresses total**.
- So instead of 4 entries, R2 can just have **1 entry** that covers **all 4 companies**.
- That one entry says: "If a packet's destination is **any address from 140.24.7.0 to 140.24.7.255**, send it to R1."
- Router R1 will then take care of forwarding it to the correct company.
- This is called **Address Aggregation** because:
- We **combine (aggregate)** several small blocks (64 IPs) into **one big block (256 IPs)**.
- This helps keep Router R2's table **short and simple**.

Address aggregation

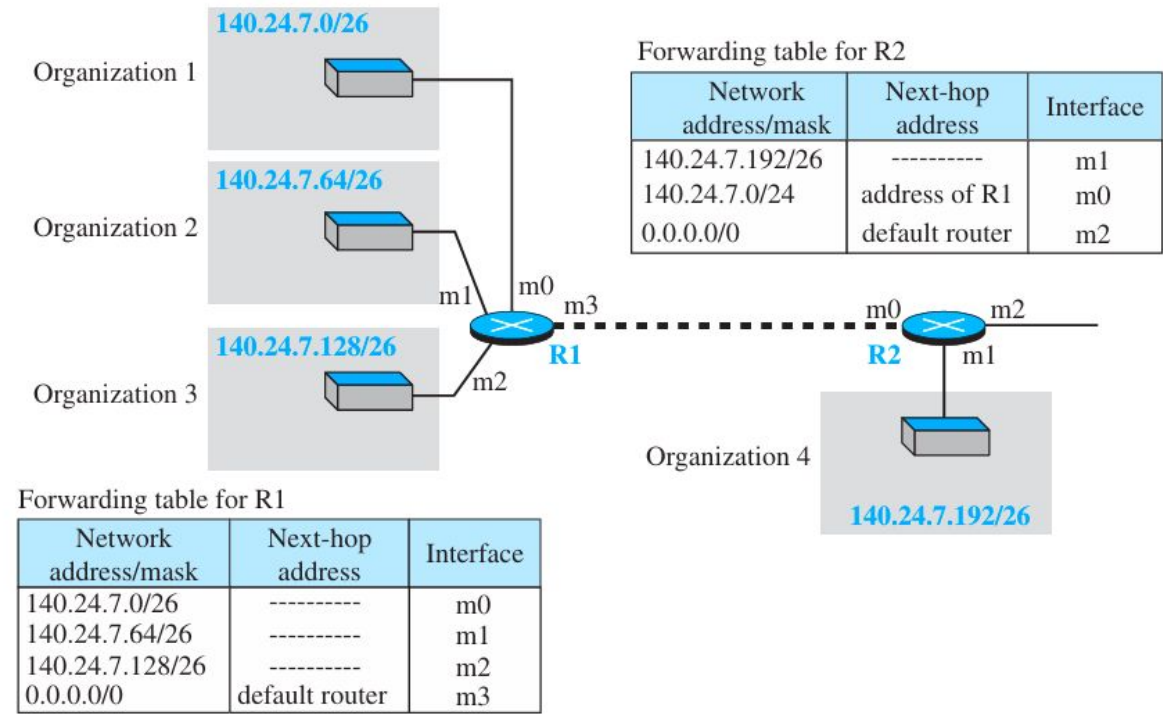
- In One Sentence:
- Router R2 doesn't care which company the packet is for — it just sends all traffic in that big block (140.24.7.0 to 140.24.7.255) to Router R1, which knows exactly where to send it.

Figure 18.34 Address aggregation



Longest Mask Matching

Figure 18.35 Longest mask matching



Suppose a packet arrives at router R2 for organization 4 with destination address 140.24.7.200.

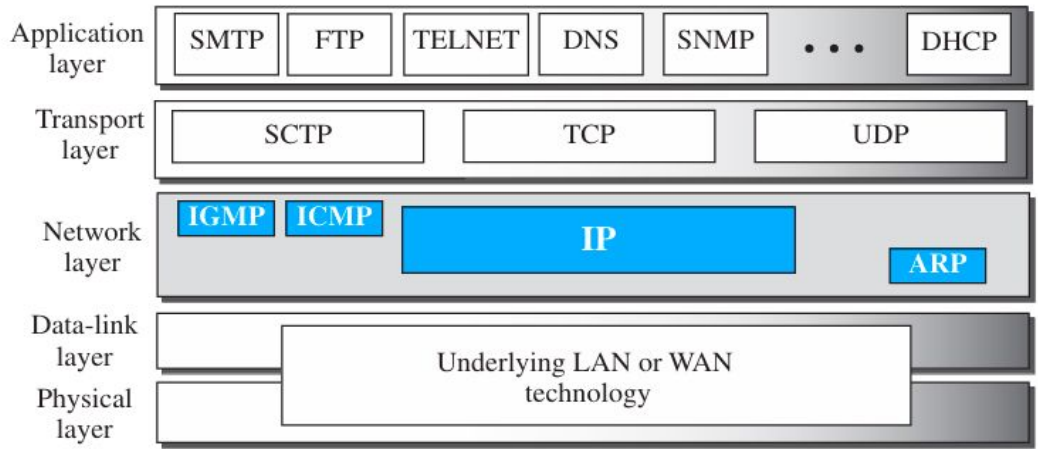
The first mask at router R2 is applied, which gives the network address 140.24.7.192. The packet is routed correctly from interface m1 and reaches organization 4.

If, however, the forwarding table was not stored with the longest prefix first, applying the /24 mask would result in the incorrect routing of the packet to router R1.

2.3. Network Layer Protocols: Internet Protocol(IP)

- The network layer in version 4 can be thought of as one main protocol and three auxiliary ones.
- The main protocol, Internet Protocol version 4 (IPv4), is responsible for packetizing, forwarding, and delivery of a packet at the network layer.
- The Internet Control Message Protocol version 4 (ICMPv4) helps IPv4 to handle some errors that may occur in the network-layer delivery.
- The Internet Group Management Protocol (IGMP) is used to help IPv4 in multicasting.
- The Address Resolution Protocol (ARP) is used to glue the network and data-link layers in mapping network-layer addresses to link-layer addresses.
- Figure 19.1 shows the positions of these four protocols in the TCP/IP protocol suite.

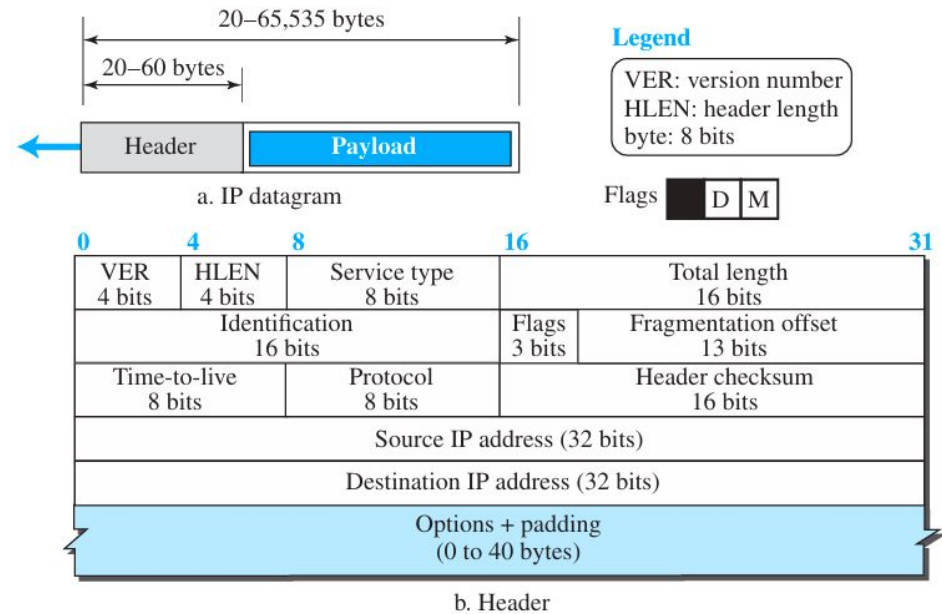
Figure 19.1 *Position of IP and other network-layer protocols in TCP/IP protocol suite*



2.3.1. IPV4 Datagram Format

- Packets in the IPv4 layer are called **datagrams**. Packets used by the IP are called **datagrams**.
- Figure 19.2 shows the IPv4 datagram format.

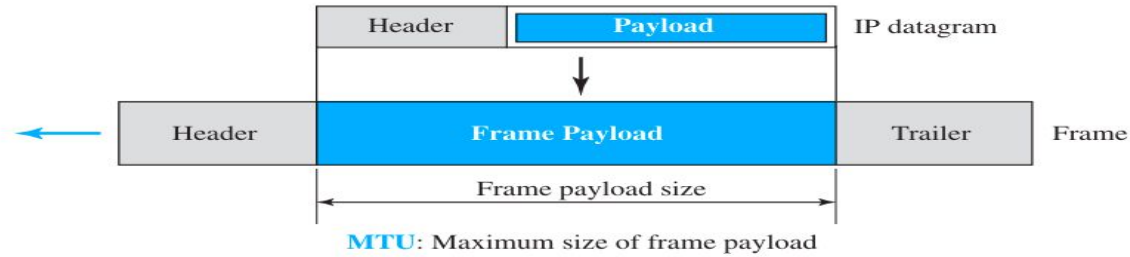
Figure 19.2 *IP datagram*



2.3.4. Fragmentation options

- Fragmentation is the process of breaking a large packet of data into smaller pieces so that it can pass through the network that supports only smaller packet sizes.
- It is needed because networks or protocols have different **MTU(Maximum Transfer Unit)**

Figure 19.5 Maximum transfer unit (MTU)

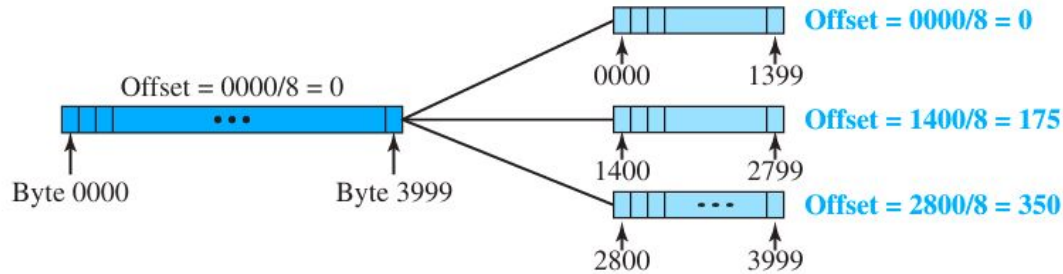


Fields Related to Fragmentation

- The fields that are related to fragmentation and reassembly of an IPv4 datagram are the **identification**, **flags**, and **fragmentation offset** fields.

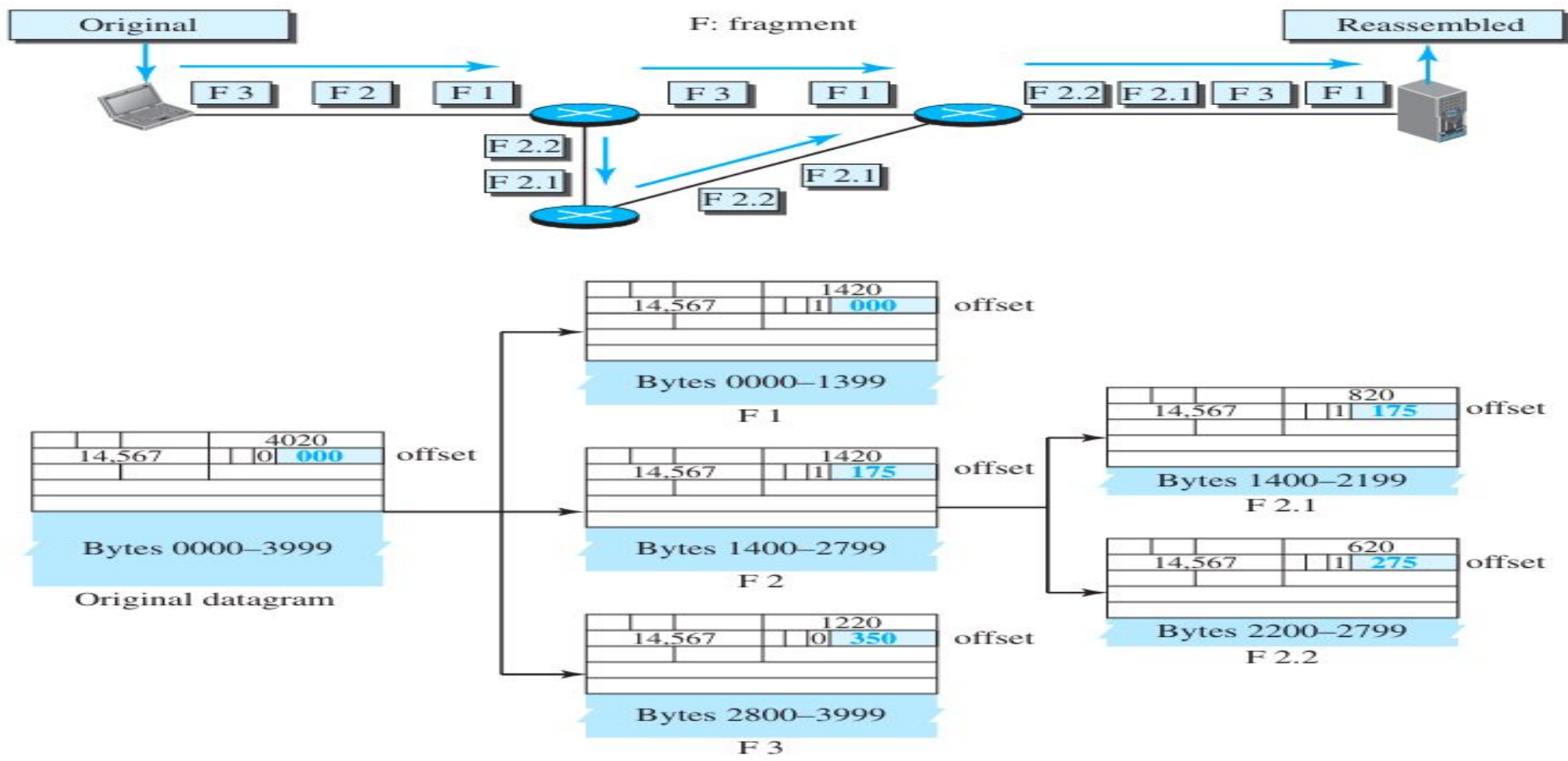
- Figure 19.6 shows a datagram with a data size of 4000 bytes fragmented into three fragments.

Figure 19.6 *Fragmentation example*



- The bytes in the original datagram are numbered 0 to 3999. The first fragment carries bytes 0 to 1399.
- The offset for this datagram is $0/8 = 0$.
- The second fragment carries bytes 1400 to 2799; the offset value for this fragment is $1400/8 = 175$.
- Finally, the third fragment carries bytes 2800 to 3999. The offset value for this fragment is $2800/8 = 350$.
- Remember that the value of the offset is measured in units of 8 bytes.

Figure 19.7 Detailed fragmentation example



Example Problem on Fragmentation

2. A Router with a MTU of 200 bytes received on IP packet of 520 bytes with an IP header of length 20 bytes what are the values relevant to fragmentation.

Example Problem on determining first middle or last fragment

3. A packet has arrived in which the offset value is 300 and payload size is 100 bytes. What are the number of First byte and Last byte?

4. A Packet has arrived in which the offset value is 100. The value of HLEn is 5 and the value of total field is 100 bytes.what are the number of first byte and last byte?

- **Example 19.6** A packet has arrived with an M bit value of 0. Is this the first fragment, the last fragment, or a middle fragment?
Do we know if the packet was fragmented?

Solution

If the M bit is 0, it means that there are no more fragments; the fragment is the last one. However, we cannot say if the original packet was fragmented or not. A non fragmented packet is considered the last fragment.

- **Example 19.7** A packet has arrived with an M bit value of 1. Is this the first fragment, the last fragment, or a middle fragment?
Do we know if the packet was fragmented?

Solution

If the M bit is 1, it means that there is at least one more fragment. This fragment can be the first one or a middle one, but not the last one. We don't know if it is the first one or a middle one; we need more information (the value of the fragmentation offset).

- **Example 19.8** A packet has arrived with an M bit value of 1 and a fragmentation offset value of 0. Is this the first fragment, the last fragment, or a middle fragment?

Solution

Because the M bit is 1, it is either the first fragment or a middle one. Because the offset value is 0, it is the first fragment.

- **Example 19.9** A packet has arrived in which the offset value is 100. What is the number of the first byte? Do we know the number of the last byte?

Solution

To find the number of the first byte, we multiply the offset value by 8. This means that the first byte number is 800. We cannot determine the number of the last byte unless we know the length of the data.

- **Example 19.10**
- A packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?

Solution

The first byte number is $100 \times 8 = 800$. The total length is 100 bytes, and the header length is 20 bytes (5×4), which means that there are 80 bytes in this datagram. If the first byte number is 800, the last byte number must be 879.

Options

- The header of the IPv4 datagram is made of two parts: a fixed part and a variable part. The fixed part is 20 bytes long and was discussed in the previous section.
- The variable part comprises the options that can be a maximum of 40 bytes (in multiples of 4-bytes) to preserve the boundary of the header.
- Options are divided into two broad categories: single-byte options and multiple-byte options.
- Options are divided into two broad categories: **single-byte options** and **multiple-byte options**.
- **Single-Byte Options**
 - **No Operation**
 - **End of Option**
- **Multiple-Byte Options**
 - **Record Route**
 - **Strict Source Route**
 - **Loose Source Route**
 - **Timestamp**

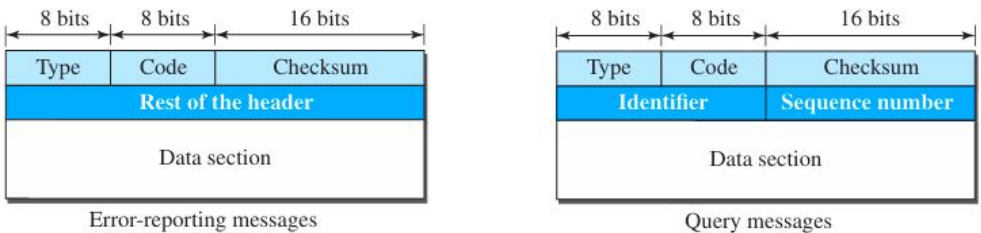
Security of IPV4 datagram

- The IPv4 protocol, as well as the whole Internet, was started when the Internet users trusted each other. No security was provided for the IPv4 protocol. Today, however, the situation is different; the Internet is not secure anymore.
- Here we discuss the security issues in IP protocol and the solutions.
- There are three security issues that are particularly applicable to the IP protocol:
 - **packet sniffing,**
 - **packet modification, and**
 - **IP spoofing.**
- **IPSec** - The IP packets today can be protected from the previously mentioned attacks using a protocol called IPSec (IP Security).
- **It Provides following services**
 - Defining Algorithms and Keys.
 - Packet Encryption.
 - Data Integrity.
 - Origin Authentication

2.4. ICMP Checksum

- The IPv4 has no error-reporting or error-correcting mechanism. What happens if something goes wrong? What happens if a router must discard a datagram because it cannot find a route to the final destination, or because the time-to-live field has a zero value?
- The IP protocol also lacks a mechanism for host and management queries. A host sometimes needs to determine if a router or another host is alive. And sometimes a network manager needs information from another host or router.
- The **Internet Control Message Protocol version 4 (ICMPv4)** has been designed to compensate for the above two deficiencies.

Figure 19.8 General format of ICMP messages



Type and code values

Error-reporting messages

- 03: Destination unreachable (codes 0 to 15)
- 04: Source quench (only code 0)
- 05: Redirection (codes 0 to 3)
- 11: Time exceeded (codes 0 and 1)
- 12: Parameter problem (codes 0 and 1)

Query messages

- 08 and 00: Echo request and reply (only code 0)
- 13 and 14: Timestamp request and reply (only code 0)

2.5. Next Generation IP

- The address depletion of IPv4 and other shortcomings of this protocol prompted a new version of IP in the early 1990s. The new version, which is called Internet Protocol version 6 (IPv6) or IP new generation (IPng) was a proposal to augment the address space of IPv4 and at the same time redesign the format of the IP packet and revise some auxiliary protocols such as ICMP.

2.5(a). IPV6 Addressing

- The main reason for migration from IPv4 to IPv6 is the small size of the address space in IPv4.
- Here we'll know how the huge address space of IPv6 prevents address depletion in the future.
- An IPv6 address is 128 bits or 16 bytes (octets) long, four times the address length in IPv4.

2.5.1(a). Representation

- A computer normally stores the address in binary, but it is clear that 128 bits cannot easily be handled by humans. Several notations have been proposed to represent IPv6 addresses when they are handled by humans.
- The following shows two of these notations: binary and colon hexadecimal.

Binary (128 bits)	1111111011110110 ... 1111111100000000
Colon Hexadecimal	FEF6:BA98:7654:3210:ADEF:BBFF:2922:FF00

- **Abbreviations**
- **Mixed Notations**
- **CIDR Notation**

2.5.2(a). Address Space

- The address space of IPv6 contains 2^{128} addresses. This address space is 2^{96} times the IPv4 address—definitely no address depletion
- To give some idea about the number of addresses, we assume that only 1/64 (almost 2 percent) of the addresses in space can be assigned to the people on planet Earth and the rest are reserved for special purposes. We also assume that the number of people on the earth is soon to be 2^{34} (more than 16 billion). Each person can have 2^{88} addresses to use. Address depletion in this version is impossible.
- **Three address types** - In IPv6, a destination address can belong to one of three categories: **unicast, anycast, and multicast**.

2.5.3(a). Address Space Allocation

- Like the address space of IPv4, the address space of IPv6 is divided into several blocks of varying size and each block is allocated for a special purpose.
- Most of the blocks are still unassigned and have been set aside for future use.
- Table 22.1 shows only the assigned blocks. In this table, the last column shows the fraction each block occupies in the whole address space.

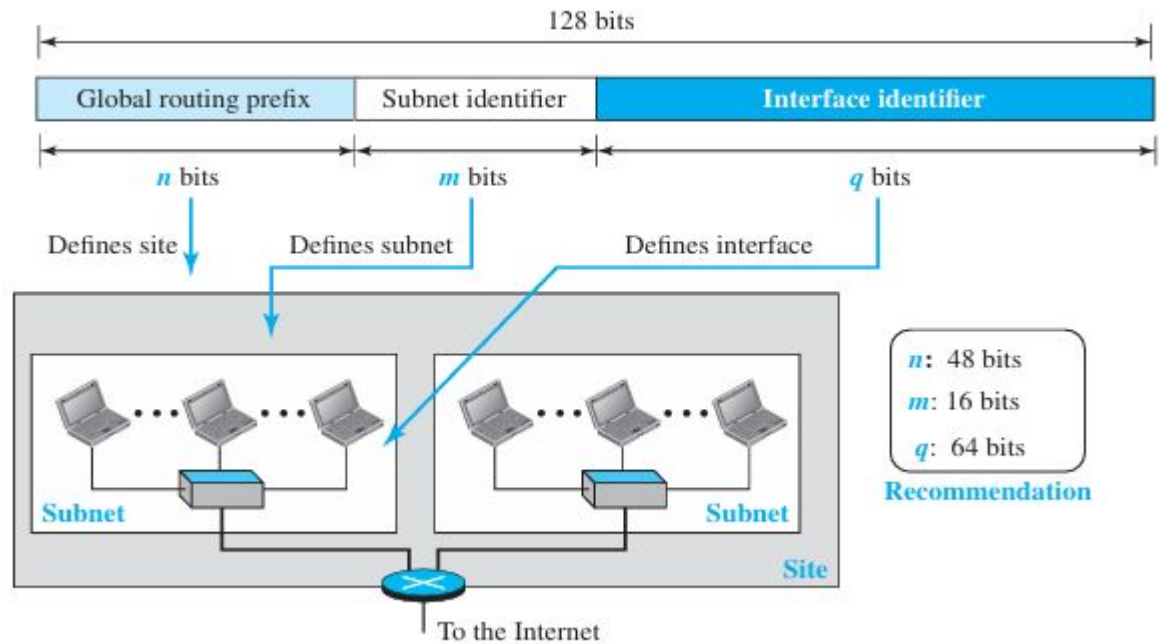
Table 22.1 *Prefixes for assigned IPv6 addresses*

<i>Block prefix</i>	<i>CIDR</i>	<i>Block assignment</i>	<i>Fraction</i>
0000 0000	0000::/8	Special addresses	1/256
001	2000::/3	Global unicast	1/8
1111 110	FC00::/7	Unique local unicast	1/128
1111 1110 10	FE80::/10	Link local addresses	1/1024
1111 1111	FF00::/8	Multicast addresses	1/256

Global unicast address

- An address in this block is divided into three parts: global routing prefix (n bits), subnet identifier (m bits), and interface identifier (q bits),

Figure 22.1 *Global unicast address*



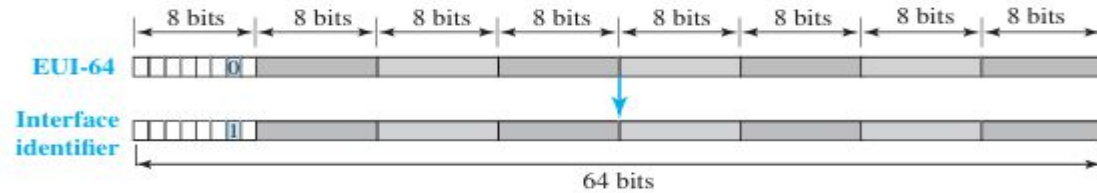
- The global routers in the Internet route a packet to its destination site based on the value of n.

- Two common link layer addressing schemes can be considered for this purpose: the 64-bit extended unique identifier (**EUI-64**) defined by IEEE and the 48-bit link-layer address defined by **Ethernet**.

Mapping EUI-64

- To map a 64-bit physical address, the global/local bit of this format needs to be changed from 0 to 1 (local to global) to define an interface address

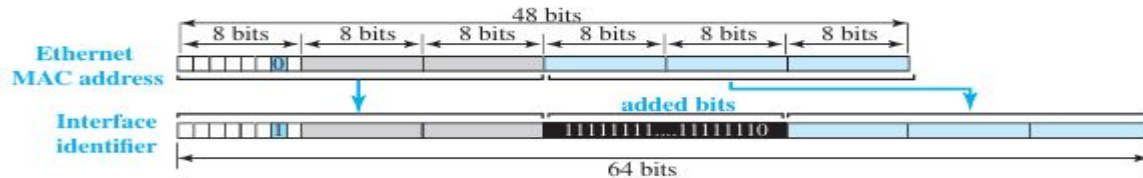
Figure 22.2 Mapping for EUI-64



Mapping Ethernet MAC Address

- Mapping a 48-bit Ethernet address into a 64-bit interface identifier is more involved. We need to change the local/global bit to 1 and insert an additional 16 bits. The additional 16 bits are defined as 15 ones followed by one zero, or FFFE16.

Figure 22.3 Mapping for Ethernet MAC



Example 1

- Using the format we defined for Ethernet addresses, find the interface identifier if the physical address in the EUI is **$(F5A9:23EF:0714:7AD2)_{16}$** .

Example 22

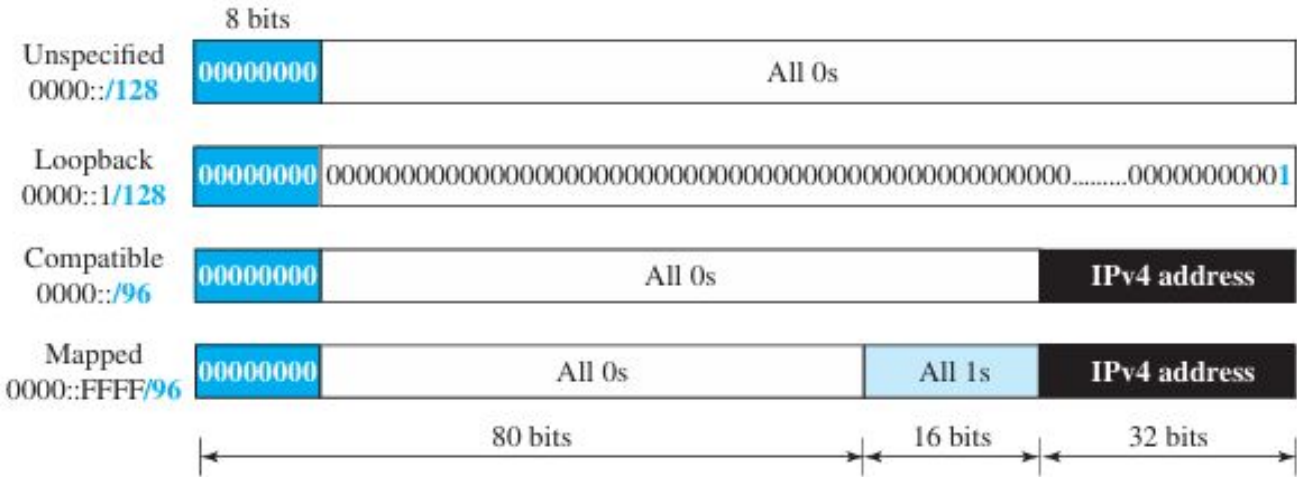
- Using the format we defined for Ethernet addresses, find the interface identifier if the Ethernet physical address is **$(F5-A9-23-14-7A-D2)_{16}$** .

Example 22.4

- An organization is assigned the block 2000:1456:2474/48. What is the IPv6 address of an interface in the third subnet if the IEEE physical address of the computer is **$(F5-A9-23-14-7A-D2)_{16}$** ?

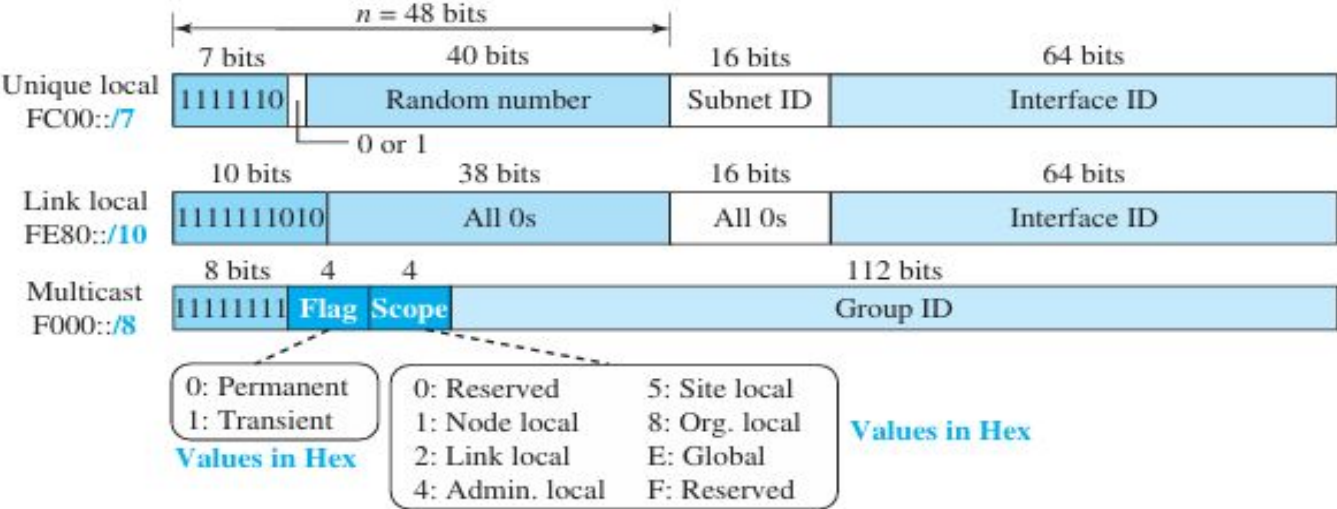
Special addresses

Figure 22.4 Special addresses



Other Assigned Blocks

Figure 22.5 Unique local unicast block



2.5.4(a). Autoconfiguration

- One of the interesting features of IPv6 addressing is the autoconfiguration of hosts. As we discussed in IPv4, the host and routers are originally configured manually by the network manager.
- However, the Dynamic Host Configuration Protocol, DHCP, can be used to allocate an IPv4 address to a host that joins the network. In IPv6, DHCP protocol can still be used to allocate an IPv6 address to a host, but a host can also configure itself.

In simple words:

1. The host creates a local address (FE80::<10 + interface ID).
2. Check if it's unique on the LAN.
3. If unique → keeps it.
Then asks the router for a global prefix.
4. Combines that prefix with its interface ID to make its Internet address.

2.5.5(a). Renumbering

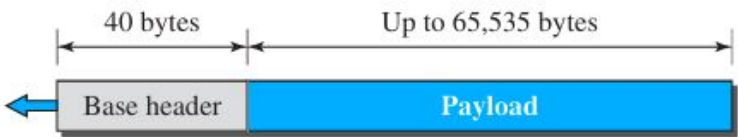
- To allow sites to change the service provider, renumbering of the address prefix (n) was built into IPv6 addressing.
- As we discussed before, each site is given a prefix by the service provider to which it is connected. If the site changes the provider, the address prefix needs to be changed.

2.5(b). The IPV6 Protocol

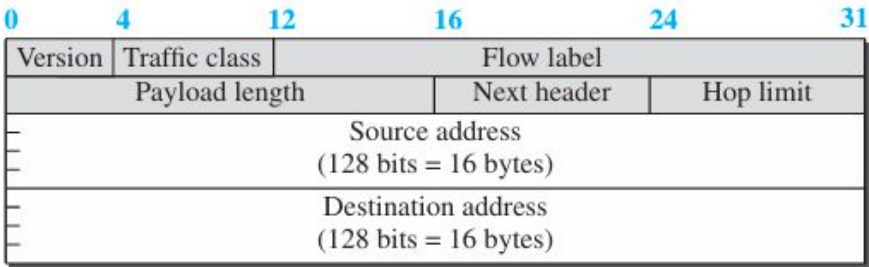
- **Better header format.**
- **New options.**
- **Allowance for extension.**
- **Support for more security.**

2.5.1(b). Packet Format

Figure 22.6 *IPv6 datagram*



a. IPv6 packet

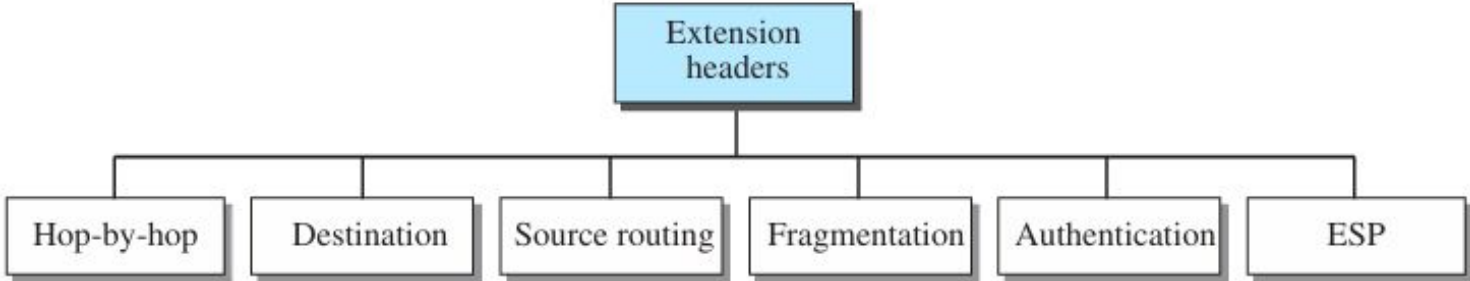


b. Base header

- **Fragmentation and Reassembly**

2.5.2(b). Extinction Header

Figure 22.8 *Extension header types*



Comparison of Options between IPv4 and IPv6

The following shows a quick comparison between the options used in IPv4 and the options used in IPv6 (as extension headers).

- The no-operation and end-of-option options in IPv4 are replaced by Pad1 and PadN options in IPv6.
- The record route option is not implemented in IPv6 because it was not used.
- The timestamp option is not implemented because it was not used.
- The source route option is called the source route extension header in IPv6.
- The fragmentation fields in the base header section of IPv4 have moved to the fragmentation extension header in IPv6.
- The authentication extension header is new in IPv6.
- The encrypted security payload extension header is new in IPv6.